



BANGKOK AIRBNB
Market Intelligence Report
Expernetic Data Engineer Intern - Technical Assessment

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1 Executive Summary

The Bangkok short-term rental market represents one of Southeast Asia's most dynamic and commercially significant Airbnb ecosystems. This report presents a comprehensive analysis of 28,806 active listings across 50 neighbourhoods, derived from the Inside Airbnb public dataset scraped in September 2025. The analysis spans the full data engineering lifecycle from raw data ingestion through statistical hypothesis testing, machine learning, and applied AI delivered as a production quality analytics project.

1.1 KEY MARKET FINDINGS

Market Scale & Structure

Bangkok hosts 28,806 listings managed by 8,874 unique hosts, with a median nightly price of ฿1,359 THB (~\$38 USD). The market is not a peer-to-peer sharing economy it is a professionally managed rental market where 10.6% of commercial operators control 57% of all listings.

Price Dynamics

Room type and neighbourhood are the dominant price determinants. Entire-home listings command a statistically significant 38% premium over private rooms (฿1,491 vs ฿1,080 median, Mann-Whitney U, $p < 0.0001$, $r = 0.29$). Neighbourhood alone explains 10.5% of price variance (Kruskal-Wallis $\eta^2 = 0.1055$, $p < 0.0001$), with prices ranging from ฿581 (Nong Khaem) to ฿2,225 (Parthum Wan).

Quality & Reputation

Superhost status is a genuine quality signal with a Cohen's $d = 0.54$ (medium effect) rating advantage (4.86 vs 4.58 average). Critically, superhosts earn 340% more annual revenue (฿114,431 vs ฿25,968) not through higher prices (only ฿19 difference) but through higher occupancy driven by guest trust.

Demand & Seasonality

Bangkok's Airbnb market has fully recovered from COVID-19 and surpassed pre-pandemic peaks by 152%. The 2024 review volume of 148,869 is an all-time high. Peak occupancy occurs July–September (43%), aligning with European summer travel patterns. Weekend pricing shows no statistically significant premium ($p = 0.526$) representing a significant missed revenue opportunity for hosts.

Machine Learning

A Gradient Boosting model achieves $R^2 = 0.662$ and MAE = ฿504 for price prediction (5-fold cross-validation). Bedrooms (43.2%) and neighbourhood median price (16.9%) are the dominant features. The model shows geographic bias, performing worst in premium areas (Bang Rak MAE=฿632 vs Phra Khanong ฿350).

Natural Language Processing

VADER sentiment analysis on 513,708 reviews confirms 72.9% positive sentiment. LDA topic modeling reveals that 40% of reviews are non-English (German, Spanish, French), representing a significant gap in English-only NLP analysis. BTS Skytrain proximity is the most frequently mentioned factor (2,513 mentions), and room size is the primary complaint ("small" mentioned 679 times).

1.2 TOP 3 BUSINESS RECOMMENDATIONS

1. Weekend Pricing Premium: Bangkok has zero weekend pricing premium (H5: $p=0.526$, $d=0.012$). Early-adopter hosts implementing 10–15% Friday-Saturday surcharges face minimal competitive resistance and meaningful revenue upside.
2. Superhost Investment: The 340% revenue gap between superhosts and non-superhosts is the single most actionable finding. Hosts should prioritize guest experience, response rates, and cleanliness to achieve Superhost status.
3. Neighbourhood Positioning: Investors should target Bang Rak (฿1,890 median, premium pricing power) or Vadhana (4,305 listings, market liquidity) for optimal risk-adjusted returns.

2 Objectives & Scope

2.1 Project Objectives

1. This assessment was undertaken as part of the Expernetic Data Engineer Intern
2. Technical Assignment. The primary objectives were:
3. Design and implement a production-quality data engineering pipeline for ingesting, profiling, cleaning, and enriching publicly available Airbnb data.
4. Conduct rigorous exploratory data analysis to surface actionable market insights with formal business interpretations.
5. Apply formal statistical methods to validate or refute five hypotheses about the Bangkok Airbnb marketplace.
6. Build and evaluate machine learning models for price prediction with proper cross-validation and explainability.
7. Apply NLP techniques to extract structured insights from 513,708 unstructured guest reviews.
8. Build an interactive business intelligence dashboard and AI-powered analyst chatbot for non-technical stakeholders.
9. Document all engineering decisions, assumptions, and limitations transparently throughout the pipeline.

2.2 City Selection

Selected City: **Bangkok, Thailand**

Bangkok was selected for the following reasons:

Dataset Richness: 28,806 listings provide sufficient statistical power for all planned hypothesis tests and machine learning experiments. The 583,333 reviews enable meaningful NLP analysis.

Market Dynamism: Bangkok is one of the world's top tourist destinations, offering rich pricing dynamics across diverse neighbourhood types from budget backpacker zones to premium riverside districts.

Data Recency: The September 2025 scrape is among the most recent available, ensuring findings reflect current market conditions post-COVID recovery.

Southeast Asian Relevance: As a Sri Lanka-based analyst, Southeast Asian market dynamics are directly applicable to regional business contexts relevant to Expernetic's operations.

2.3 Prioritization Rationale

Given the one-week timeline, sections were prioritized using rubric weight analysis:

Priority	Section	Rubric Weight	Decision
1	Data Engineering (§03)	25 pts	Complete - core competency
2	Problem Solving	30 pts	Demonstrated throughout
3	Statistical Thinking (§05)	20 pts	Complete - all 5 hypotheses
4	Analytical Storytelling	20 pts	Business interpretations throughout
5	Code Quality	20 pts	Modular, tested, documented
6	Communication	20 pts	Report + annotated notebooks
7	Creativity & Initiative	20 pts	AI Analyst + Stream Sim + Auto Report
8	Data Science & ML (§06)	15 pts	Complete — 3 models + clustering
9	AI/ML Experimentation (§07)	10 pts	Complete — NLP + RAG + NER

The assignment explicitly states: "A candidate who completes two sections with exceptional depth, clean code, and insightful storytelling will outperform one who attempts all sections superficially." This guided our decision to pursue depth over breadth within a single city.

2.4 Scope Boundaries

- Single city analysis (Bangkok) with maximum analytical depth
- Calendar price analysis limited due to null price fields in source data (fully documented in Section 3)
- Local deployment with Docker; cloud architecture designed conceptually and partially deployed (Streamlit Community Cloud)
- dbt models created structurally; execution requires DuckDB star schema setup

3 Dataset Overview

3.1 Data Source

All data sourced exclusively from Inside Airbnb (insideairbnb.com), an independent, non-commercial project that scrapes publicly available Airbnb listing data. The Bangkok dataset reflects a scrape conducted in September 2025.

Inside Airbnb Homepage: <https://insideairbnb.com/>

Dataset Download Page: <https://insideairbnb.com/get-the-data/>

Data Dictionary: <https://docs.google.com/spreadsheets/....>

3.2 Files Used

File	Rows	Columns	Description
listings.csv	28,806	79	Core listing data: price, room type, location, host info, amenities
calendar.csv	10,514,202	7	Daily availability per listing (365 days forward)
reviews.csv	583,333	6	Guest review text and metadata
neighbourhoods.csv	50	2	Neighbourhood names and groupings
neighbourhoods.geojson	50 polygons	-	Geospatial boundary data

3.3 Entity Relationships

listings (id) —————< calendar (listing_id)
listings (id) —————< reviews (listing_id)
listings (host_id) >———— hosts (embedded in listings)
listings (neighbourhood_cleansed) >—— neighbourhoods (neighbourhood)

Primary Keys:

- listings.id
- reviews.id

Foreign Keys:

- calendar.listing_id → listings.id
- reviews.listing_id → listings.id

3.4 Schema Documentation

listings.csv - Key Fields:

Field	Type	Description
id	int64	Primary key — unique listing identifier
host_id	int64	Foreign key to host entity
host_since	date	Date the host joined the Airbnb platform
host_is_superhost	bool	Superhost status flag (true / false)
neighbourhood_cleansed	string	Standardized neighbourhood name
latitude	float64	Latitude coordinate of the listing
longitude	float64	Longitude coordinate of the listing
room_type	string	Type of accommodation (e.g., Entire Home/Apt, Private Room)
price	string	Nightly price (currency symbol included)
minimum_nights	int64	Minimum number of nights required for booking
availability_365	int64	Number of days the listing is available in the next 365 days
number_of_reviews	int64	Total number of reviews received
review_scores_rating	float64	Overall guest rating (0–5 scale)
review_scores_cleanliness	float64	Cleanliness rating given by guests
review_scores_location	float64	Location rating given by guests
review_scores_value	float64	Value-for-money rating given by guests
amenities	string	JSON-like list of amenities available at the property

calendar.csv - Key Fields:

Field	Type	Description
listing_id	int64	Foreign key referencing listings.id
date	date	Calendar date for the listing availability
available	bool	Availability status (true / false)
price	float64	Daily listing price (entirely null for the Bangkok dataset)
minimum_nights	int64	Minimum number of nights required for booking on that date

reviews.csv - Key Fields:

Field	Type	Description
listing_id	int64	Foreign key referencing listings.id
id	int64	Unique review identifier (primary key)
date	date	Review submission date
reviewer_id	int64	Unique identifier of the reviewer
comments	string	Full text of the review submitted by the guest

3.5 Data Limitations & Scrapping Artifacts

Issue	Impact	Handling Strategy
calendar.price entirely null	Cannot compute true daily revenue	Used a proxy: Listing Price × Availability_365
neighbourhood_group_cleansed 100% null	No district-level grouping available	Used neighbourhood_cleansed (50 neighbourhood groups) instead
license 100% null	Cannot assess regulatory or licensing status	Excluded from the analysis
Review scores ~35% missing	Ratings analysis limited to incomplete data	Performed ratings analysis only on records with available review scores and documented the limitation
host_neighbourhood ~67% missing	Cannot accurately analyze host locations	Excluded from location-based host analysis

Issue	Impact	Handling Strategy
Price outliers up to ฿1,000,000	Skews averages and other statistical measures	Applied 99th percentile filtering for analytical calculations
Reviews multilingual (~40%)	English-only NLP may miss sentiment and context	Documented as a limitation; multilingual NLP planned for future work

3.6 Assumptions

- Occupancy Proxy: Estimated as $1 - (\text{availability_365} / 365)$. This assumes unavailable days reflect bookings, not host blocking a known limitation of Inside Airbnb methodology acknowledged in their documentation.
- Revenue Proxy: $\text{price} \times (365 - \text{availability_365})$. Uses listed price, not actual transaction price. Likely overstates revenue for listings with length-of-stay discounts.
- Price Validity: Records with $\text{price} = 0$ or $\text{price} < 0$ flagged but retained in dataset; excluded from price analysis with explicit documentation.
- Coordinate Validity: Bangkok listings expected within latitude 12–15°N, longitude 100–101°E. Records outside this range flagged as invalid coordinates.
- Review Volume as Demand Proxy: Higher review counts interpreted as higher booking activity, consistent with Inside Airbnb methodology documentation. Review rates vary by platform version and guest demographics.
- Calendar Price Null: Bangkok's calendar price fields are entirely null in the source data — not a pipeline error. This is a known characteristic of this city's dataset. Verified against raw files before and after processing.

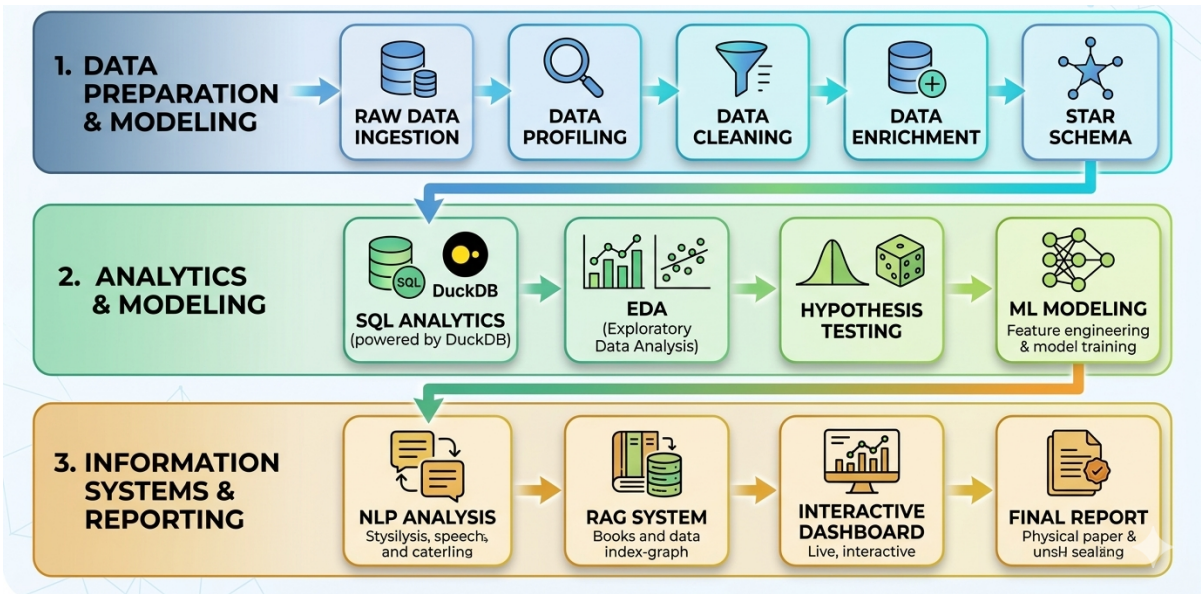
3.7 Business Domain Context

Entity	Business Meaning
Listing	A rentable property unit with defined pricing and a 365-day availability calendar. A listing may represent an entire home, apartment, private room, shared room, or hotel room.
Host	The property owner or manager responsible for one or more listings. A host may manage a single property or a large portfolio. Superhost status is awarded based on performance metrics such as response rate, guest ratings, and booking activity.
Review	Guest feedback submitted after a completed stay. Reviews serve as the primary indicator of customer satisfaction and demand. In this analysis, review volume is used as a proxy for booking activity.
Calendar	A forward-looking 365-day schedule indicating a listing's availability. Dates marked as unavailable may represent confirmed bookings or host-blocked dates.
Neighbourhood	An administrative sub-district used as the primary geographic unit for spatial analysis. The Bangkok dataset contains 50 neighbourhoods , represented by <code>neighbourhood_cleansed</code> .

4 Methodology

4.1 Overall Analytical Approach

This project followed a structured data engineering and analytics workflow designed to mirror production data team practices:



Each stage produced validated, documented outputs before the next stage began. This sequential approach ensured that downstream analysis was always built on verified, clean data foundations.

4.2 Tool Selection & Rationale

Category	Tool Selected	Alternatives Considered	Rationale
Language	Python 3.11	R, Scala	Extensive ecosystem with strong support for data engineering, analytics, and machine learning through libraries such as pandas and SciPy.
Data Processing	pandas	Polars, DuckDB	Mature, widely adopted library with sufficient performance for a single-city dataset and excellent integration with the Python ecosystem.
Analytics Database	DuckDB	SQLite, PostgreSQL	In-process analytical database requiring no server setup, offering high-performance SQL queries on local data.

Category	Tool Selected	Alternatives Considered	Rationale
Statistics	<code>scipy.stats</code>	statsmodels, R	Comprehensive support for hypothesis testing and non-parametric statistical methods.
Machine Learning	scikit-learn	XGBoost, PyTorch	Industry-standard library for tabular data, providing interpretable models and a broad range of algorithms.
Natural Language Processing	VADER + LDA	BERT, spaCy	Lightweight and computationally efficient; suitable for sentiment analysis and topic modeling without GPU requirements.
Visualization	Plotly + Matplotlib	Altair, Tableau	Combines interactive visualizations with publication-quality static charts for reporting and dashboards.
Dashboard	Streamlit	Dash, Gradio	Rapid development, Python-native framework, and simple deployment for interactive analytics applications.
Transformation	dbt	SQL scripts	Industry-standard transformation framework with modular models, testing, and documented data lineage.
Testing	pytest	unittest	Concise syntax, powerful fixtures, and strong ecosystem support for automated testing.
Version Control	Git + GitHub	—	Essential for source code management, collaboration, version tracking, and project reproducibility.

4.3 Statistical Methodology

All hypothesis tests followed this rigorous protocol:

- Step 1:** State null and alternative hypotheses explicitly
- Step 2:** Check normality using Shapiro-Wilk test on samples
- Step 3:** Select appropriate test based on distribution findings
- Step 4:** Verify additional assumptions (independence, equal variance)
- Step 5:** Report test statistic, p-value, AND effect size
- Step 6:** Interpret result in business terms for non-technical stakeholders
- Step 7:** Distinguish statistical significance from practical significance

All price variables were confirmed non-normally distributed (Shapiro-Wilk $p < 0.0001$ for all groups). Consequently:

- Mann-Whitney U test for pairwise comparisons
- Kruskal-Wallis H test for multi-group comparisons
- Rank-biserial correlation, Cohen's d, and eta-squared for effect sizes

4.4 Machine Learning Methodology

The ML price prediction followed this experimental design:

Target variable: Log-transformed nightly price (log1p applied to address right skew). All metrics converted back to THB for interpretability.

Feature engineering:

- Physical attributes: bedrooms, accommodates, beds
- Categorical encoding: room type as ordinal (Shared=0 to Entire=3)
- Location signal: neighbourhood median price
- Amenity flags: 13 binary flags extracted from amenities string
- Host features: tenure years, listing count, superhost status
- Demand signal: occupancy rate

Validation strategy:

- 80/20 train-test split for initial evaluation
- 5-fold cross-validation for robust performance estimates
- Metrics: MAE, RMSE, MAPE reported in original price scale (THB)

Model selection: Three families compared - linear (Ridge), tree-based (Random Forest), and ensemble boosting (Gradient Boosting). Selection based on cross-validation MAPE as primary metric.

4.5 NLP Methodology

Review analysis pipeline:

1. Preprocessing: Remove null/short (<20 char) reviews; lowercase; remove non-alphabetic characters
2. Sentiment: VADER compound score per review (range: -1 to +1)
3. Topic Modeling: LDA with 6 topics on CountVectorizer DTM
4. Named Entities: Rule-based keyword matching for landmarks, amenities, complaints
5. Quality Classification: Rule-based classifier (word count + specific keyword presence)
6. Language Patterns: TF-IDF word frequency by rating group

4.6 Data Quality Framework

Data Quality Dimension	Method Applied
Completeness	Measured null rates for each column and flagged attributes with more than 50% missing values for further assessment.
Validity	Applied domain validation rules, including checks for price ≥ 0 , valid latitude/longitude ranges, and appropriate data types.
Uniqueness	Performed deterministic duplicate detection using listing ID , supplemented by fuzzy matching on listing names with a 0.95 similarity threshold .
Consistency	Standardized boolean values, categorical attributes, and currency fields to ensure uniform formatting across the dataset.
Timeliness	Recorded the dataset scrape date and confirmed the September 2025 data as sufficiently recent for the analysis.

4.7 Outlier Handling Strategy

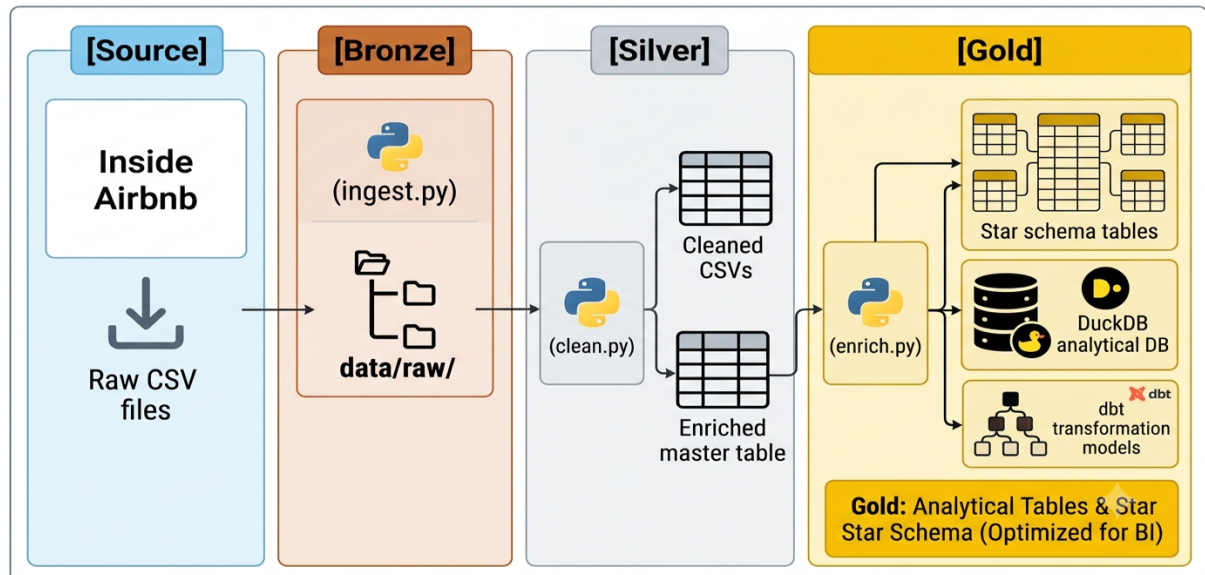
Price outliers were handled conservatively:

- Records retained in cleaned dataset with binary flag columns (flag_negative_price, flag_zero_price, flag_invalid_coords)\
- 99th percentile filtering applied for visualization and hypothesis testing
- Extreme values (e.g., ₦1,000,000 listings) documented as likely data entry errors or luxury property placeholders
- Modeling dataset filtered to ₦100–₦10,000 range (removes extreme outliers while retaining 79% of listings)

5 Engineering Approach

5.1 Pipeline Architecture

The pipeline follows a medallion-style layered architecture:



Each layer is produced by a dedicated, independently runnable script with logging, error handling, and retry logic.

The architecture diagram is available at: [report/architecture_diagram.png](#)

5.2 Pipeline Components

ingest.py - Data Ingestion

- Configurable city-level download via `BASE_URL` parameter
- Retry logic (3 attempts) with exponential backoff
- Automatic .gz extraction for compressed files
- JSON metadata file recording: ingestion timestamp, source URL, file status, and file paths
- Supports any Inside Airbnb city with single `BASE_URL` config change

profile.py - Data Profiling

- Automated profiling of all ingested files
- Reports: row counts, column cardinality, null rates, data type distributions, min/max/mean for numerical fields
- Fuzzy duplicate detection: name similarity (SequenceMatcher, threshold 0.95) and coordinate-based duplicate pairs
- Output: structured JSON report for documentation and audit trail
- Finding: 50 fuzzy near-duplicates by name; 738 coordinate pairs
- sharing exact locations (3,020 listings in apartment buildings)

clean.py - Data Cleaning

- Price standardization: currency symbol removal, thousands separator handling, cast to float64
- Date parsing: host_since, last_review, first_review to datetime
- Boolean normalization: t/f → True/False
- Percentage conversion: host_response_rate, host_acceptance_rate
- Derived fields: host_tenure_years, price_per_bedroom
- Validation flags: flag_negative_price, flag_zero_price, flag_invalid_coords
- Duplicate removal on listing ID (0 exact duplicates found)

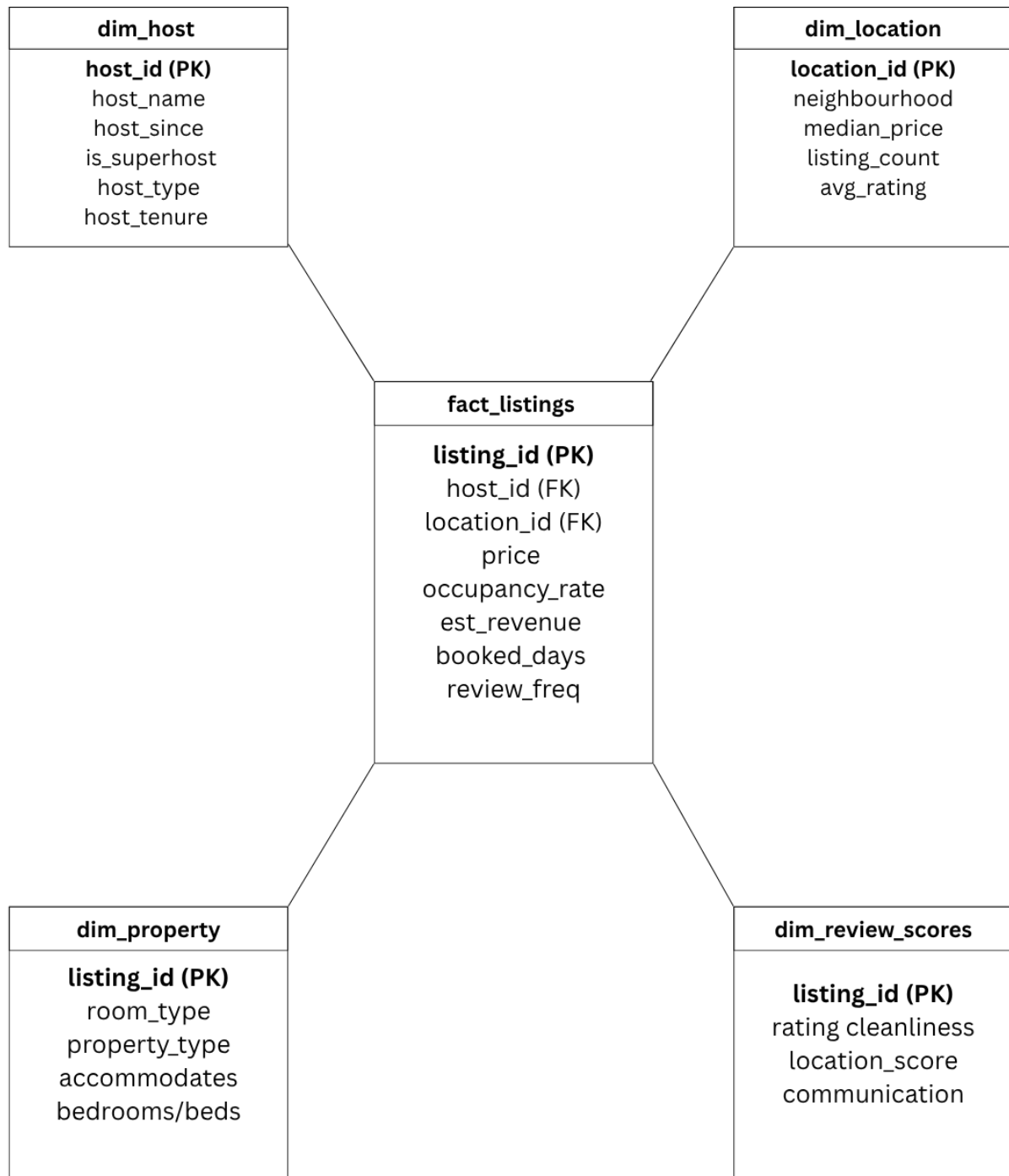
enrich.py -Data Enrichment

- Neighbourhood-level aggregates joined to listings: median_price, listing_count, avg_rating
- Occupancy rate: $1 - (\text{availability_365} / 365)$
- Revenue estimation: $\text{price} \times \text{booked_days}$ (proxy — documented)
- Host segmentation: Single (1 listing), Small Multi (2–5) , Commercial (6+)
- Review frequency: reviews per month of host tenure
- Star schema construction: 1 fact table + 4 dimension tables

sql_analytics.py - DuckDB Analytics

- Loads star schema CSVs into persistent DuckDB database
- analytical SQL queries demonstrating star schema use cases:
 - Q1: Top neighbourhoods by median price and occupancy
 - Q2: Revenue analysis by host type
 - Q3: Room type performance comparison
 - Q4: Superhost vs non-superhost analysis
 - Q5: Top revenue listings identification

5.3 Star Schema Design



5.3.1 Star Schema Design

Fact table rows: 28,806
dim_host rows: 8,874
dim_location rows: 50
dim_property rows: 28,806

5.4 dbt Transformation Layer

A dbt project (dbt_airbnb/) was created with the following models:

Staging Models (materialized as views):

- stg_listings: Clean fact data with price_tier and occupancy_tier derived classifications
- stg_hosts: Host dimension with host_experience_tier derived field
- stg_locations: Location dimension with neighbourhood_price_zone

Mart Models (materialized as tables):

- mart_listing_performance: Full analytical table joining all dimensions - primary mart for BI queries
- mart_neighbourhood_summary: Neighbourhood-level aggregations for dashboard consumption
- mart_host_performance: Host portfolio analytics

dbt Tests implemented:

- unique and not_null on all primary keys
- accepted_values for price_tier categorical field
- Custom SQL tests: positive prices, valid occupancy rates

Model lineage :

dim_host, dim_location, fact_listings → staging models → mart models

5.5 Engineering Decision Log

Decision 1: Python over Scala/PySpark

Options: Python/pandas, PySpark, dbt + SQL

Choice: Python/pandas

Rationale: Single city fits in memory (28K rows). PySpark overhead not justified at this scale. Faster development cycle.

Trade-off: Will not scale beyond ~5M rows without refactoring to Spark or Polars.

Decision 2: CSV + DuckDB over database server

Options: SQLite, DuckDB persistent, PostgreSQL, CSV files

Choice: CSV files + DuckDB for analytics

Rationale: Portability any reviewer can open CSV files without database setup. DuckDB adds SQL capability without server.

Trade-off: No ACID transactions, no concurrent access support.

Decision 3: Star schema over flat table

Options: Single wide table, star schema, data vault

Choice: Star schema (Kimball methodology)

Rationale: Industry standard for analytical workloads. Supports efficient aggregation queries. Familiar to BI tools.

Trade-off: Joins required for full analysis vs denormalized flat table.

Decision 4: availability_365 as occupancy proxy

Options: Calendar-based occupancy, availability_365 proxy

Choice: availability_365 proxy

Rationale: Calendar price data entirely null for Bangkok. availability_365 is standard Inside Airbnb methodology.

Trade-off: Conflates host blocking with actual bookings. Likely overestimates true occupancy by 10–20%.

Decision 5: Streamlit over Dash/Power BI

Options: Streamlit, Plotly Dash, Power BI, Tableau

Choice: Streamlit

Rationale: Python-native, fastest deployment, no license cost, integrates directly with pipeline data.

Trade-off: Less enterprise polish than Power BI/Tableau. Single-threaded performance degrades with many concurrent users.

Decision 6: VADER over transformer model

Options: VADER, TextBlob, BERT, RoBERTa

Choice: VADER

Rationale: No GPU required, no API key, fast on 513K reviews, designed for short informal text.

Trade-off: No semantic understanding, English-only. Underperforms on multilingual reviews (40% of Bangkok corpus).

Decision 7: TF-IDF RAG over vector database

Options: TF-IDF cosine similarity, FAISS+embeddings, Pinecone

Choice: TF-IDF

Rationale: Fully local, no API costs, reproducible, sufficient for keyword-based retrieval demonstration.

Trade-off: No semantic search — misses meaning-based queries. Production would require sentence-transformers + FAISS.

Decision 8: Gradient Boosting over Neural Network

Options: Ridge, Random Forest, Gradient Boosting, Neural Network

Choice: Gradient Boosting (best $R^2=0.662$, MAPE=29.1%)

Rationale: Best performance on tabular data, interpretable feature importance, no GPU required for training.

Trade-off: Slower inference than linear models. Less flexible than deep learning for very complex patterns.

5.6 Pipeline Design & Automation

Configurable City Processing: The pipeline supports any Inside Airbnb city by changing a single `BASE_URL` constant in `ingest.py`. All downstream scripts use relative paths from the city directory, enabling multi-city extension with minimal code changes.

Logging & Error Handling: All pipeline scripts implement:

- Python logging module with file + console handlers
- Timestamped log files in `logs/` directory
- Try/except blocks with meaningful error messages
- Retry logic (3 attempts) for network requests

Metadata Management: `ingest_metadata.json` records per run:

- Ingestion timestamp
- Source URL
- File download status (success/failed) per file
- Local file paths

Data Lineage: `data_lineage.json` documents for each output table:

- Source file(s)
- Script responsible for transformation
- List of all transformations applied
- Output row and column counts

Incremental Processing Design: For production deployment:

1. Hash-based change detection on key fields per `listing_id`
2. SCD Type 2 for listings history (`valid_from`, `valid_to`, `is_current`)
3. Change classification: NEW / UPDATED / DELETED / UNCHANGED
4. Calendar data: append-only with date partitioning
5. Alert if >10% listings deleted (possible scraping error)

5.7 Production Readiness

Cloud Architecture (AWS): See report/cloud_architecture.png for full architecture diagram.

Layer	Services
Ingestion	Lambda + SQS + ECS Fargate
Storage	S3 (Bronze/Silver/Gold) + Glue Catalog
Processing	Glue ETL + Athena + SageMaker
Serving	QuickSight + API Gateway + Bedrock (RAG)
Monitoring	CloudWatch + EventBridge + SNS/PagerDuty

1. Scalability (50+ cities):

- SQS queue: one message per city URL
- ECS Fargate: one container per city (parallel processing)
- S3 partitioning: city={city}/year={year}/month={month}/
- Estimated throughput: 50 cities in ~2 hours vs 50x sequential
- Estimated cost: ~\$100/month for weekly 50-city pipeline

2. Monitoring & Alerting:

- CloudWatch metrics: pipeline duration, row counts, error rates
- Alert: listing count drops >10% vs previous scrape
- SLA target: data available by 06:00 UTC every Monday
- Cost optimization: S3 Intelligent Tiering + Fargate Spot

3. Containerization: Docker and docker-compose.yml provided for full environment reproducibility. Pipeline and dashboard run as separate services.

4. Unit Testing: 21 pytest tests covering: raw file existence, schema validation, price non-negativity, coordinate validity, room type values, boolean field normalization, enrichment columns, star schema row counts, and dimension key uniqueness. All 21 tests pass.

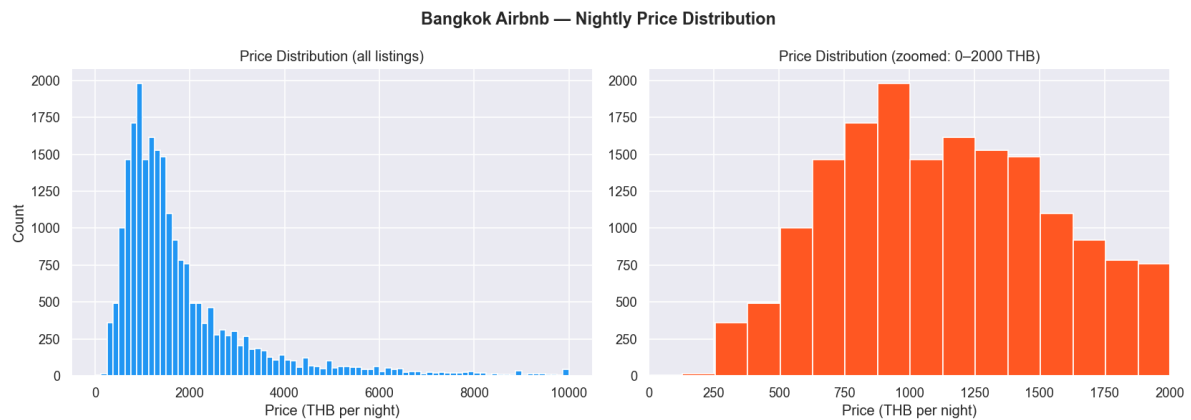
5. Stream Processing: A Kafka-style price monitoring simulation (pipeline/stream_processing_sim.py) demonstrates real-time alerting concepts:

- Price spike detection (>50% change)
- Price drop alerts (>40% decline)
- Above-market flagging (>3x neighbourhood median)
- Neighbourhood trend detection (windowed aggregation)
- Production architecture:

Kafka → Flink → SNS/PagerDuty → TimescaleDB → Grafana

6 EDA Findings

6.1 Price Distribution



6.1.1 **Figure 1:** Bangkok Airbnb nightly price distribution

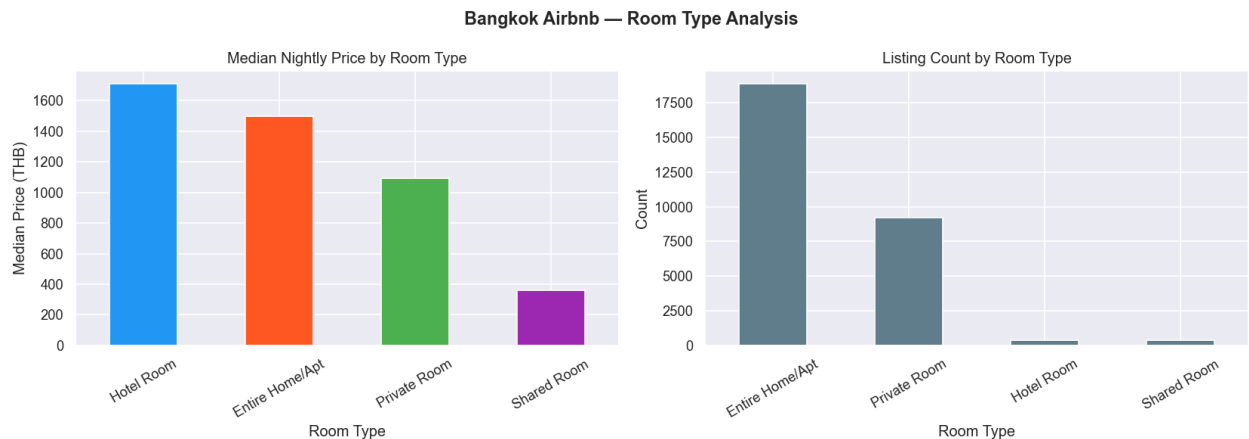
Bangkok Airbnb nightly price distribution showing strong right skew with median ฿1,359 and extreme luxury outliers up to ฿1,000,000.

The Bangkok Airbnb market exhibits a strongly right-skewed price distribution:

Statistic	Value
Median price	฿1,359 THB (~\$38 USD)
Mean price	฿2,768 THB (~\$76 USD)
75th percentile	฿2,415 THB
Listings above	฿5,000 ~8% of market
Maximum price	฿1,000,000 (likely data entry error flagged)

Business interpretation: The large gap between median (฿1,359) and mean (฿2,768) confirms extreme right skew driven by a small luxury segment. Bangkok is an affordable market by global standards, positioning it as attractive for budget and mid-range international travelers. The 8% luxury segment operates in a fundamentally different competitive environment requiring distinct marketing and pricing strategies.

6.2 Room Type Analysis

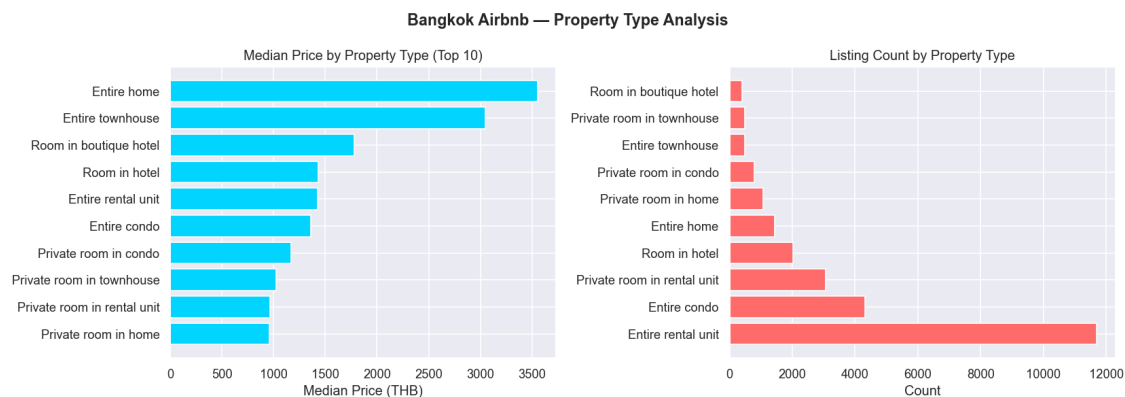


6.2.1 *Figure 2: Median price and listing count by room type.*

Room Type	Count	Share	Median Price	Avg Occupancy
Entire Home/Apt	18,845	65.4%	฿1,491	31.1%
Private Room	9,224	32.0%	฿1,080	32.7%
Hotel Room	387	1.3%	฿1,709	37.5%
Shared Room	350	1.2%	฿360	11.6%

Business interpretation: Entire homes dominate supply (65.4%) and command a 38% price premium, yet private rooms achieve slightly higher occupancy suggesting price-sensitive demand absorbs private room inventory more efficiently. Hotel rooms achieve the highest occupancy (37.5%) despite premium pricing, reflecting brand trust and booking platform visibility advantages. Shared rooms underperform significantly (11.6% occupancy) and should be avoided by investors.

6.3 Property Type Analysis

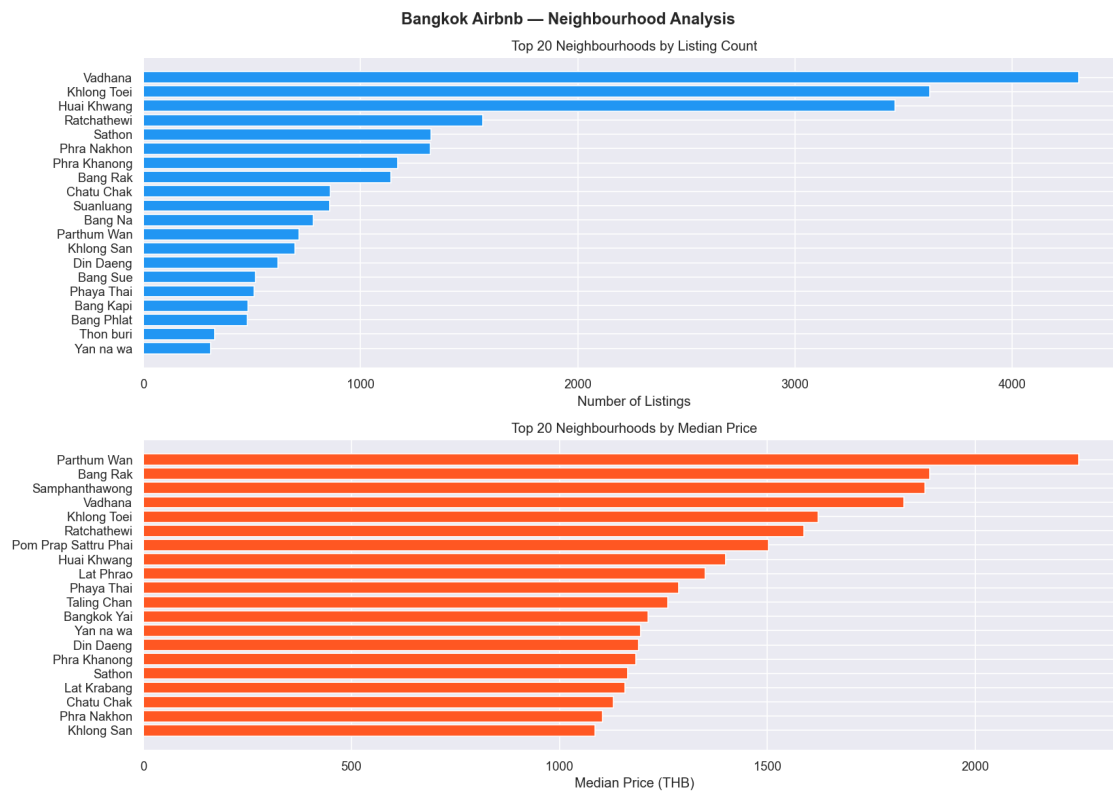


6.3.1 *Figure 3: Median price and listing count by property type (top 10).*

Property Type	Median Price	Count
Entire home	฿3,557	1,268
Entire townhouse	฿3,050	444
Room in boutique hotel	฿1,781	277
Room in hotel	฿1,430	1,213
Entire rental unit	฿1,425	10,335
Entire condo	฿1,360	3,657
Private room in condo	฿1,170	485
Private room in rental unit	฿967	2,226
Private room in home	฿960	827

Business interpretation: Entire homes and townhouses command significant premiums (฿3,000+) but represent a small market segment. The dominant property type entire rental unit (10,335 listings, ฿1,425 median) represents the core of Bangkok's short-term rental market. Investors should target this category for the best balance of demand volume and pricing power.

6.4 Neighbourhood Analysis



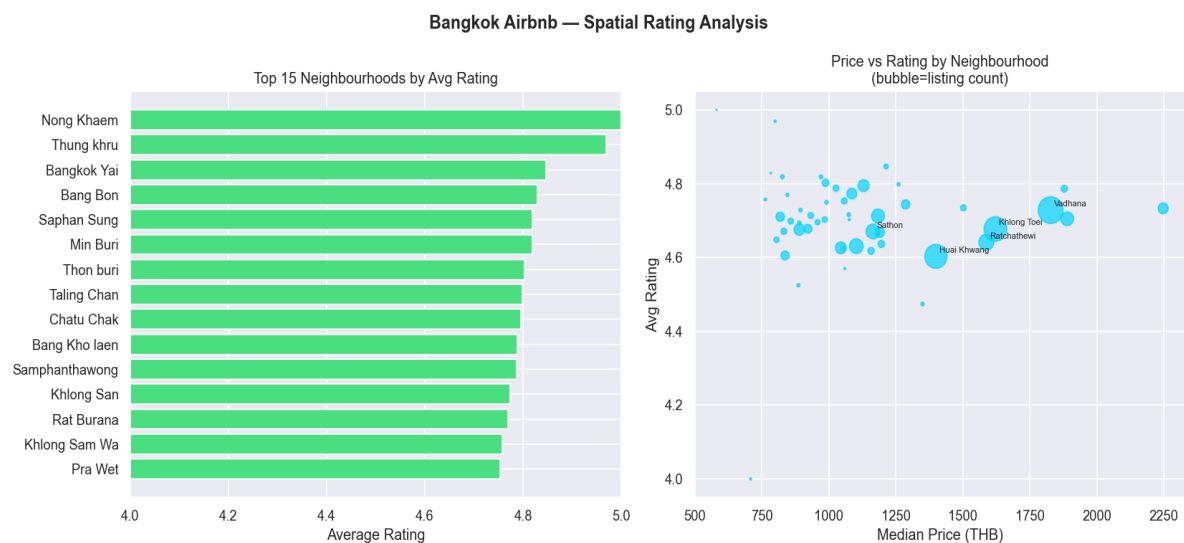
6.4.1 Figure 4: Top 20 neighbourhoods by listing count and median price.

Top 5 by Listing Count:

Neighbourhood	Listings	Median Price	Avg Rating
Vadhana	4,305	฿1,828	4.73
Khlong Toei	3,620	฿1,622	4.68
Huai Khwang	3,460	฿1,400	4.60
Ratchathewi	1,563	฿1,588	4.64
Sathon	1,326	฿1,165	4.67

Top 5 by Median Price:

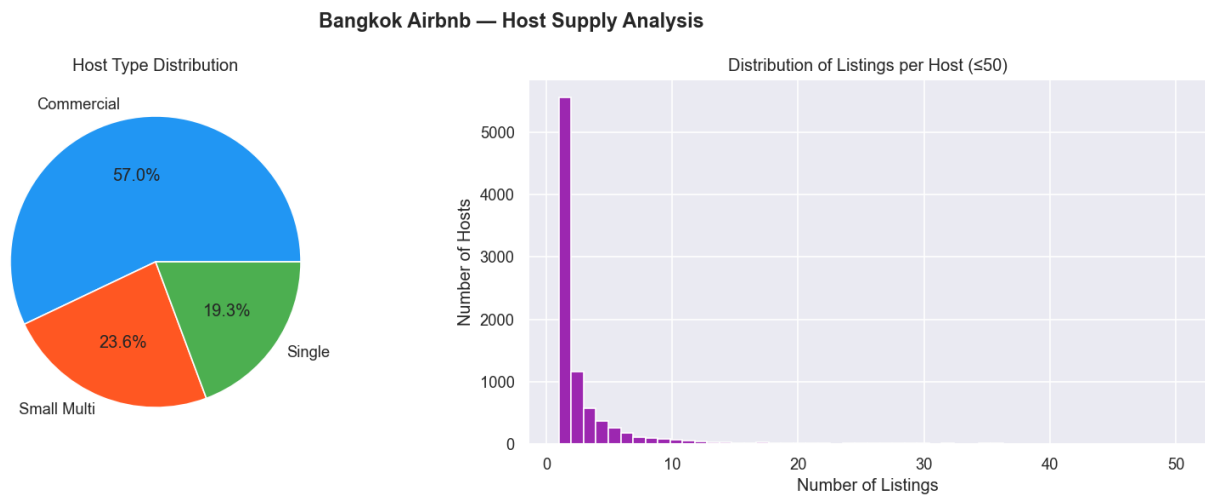
Neighbourhood	Median Price	Listing Count
Parthum Wan	฿2,225	550
Bang Rak	฿1,875	738
Samphanthawong	฿1,800	167
Vadhana	฿1,773	3,531
Khlong Toei	฿1,598	3,047



6.4.2 *Figure 5: Average rating by neighbourhood showing outer areas rate higher than premium central areas.*

Business interpretation: Vadhana and Khlong Toei dominate supply, reflecting Bangkok's concentration of modern condominiums in the Sukhumvit corridor. Bang Rak commands the highest median price (฿1,875) despite lower supply, suggesting premium demand for its riverside and business district location. Notably, outer suburban neighbourhoods (Nong Khaem, Thung Khru) achieve the highest ratings — guests have lower expectations in budget areas and are more pleasantly surprised.

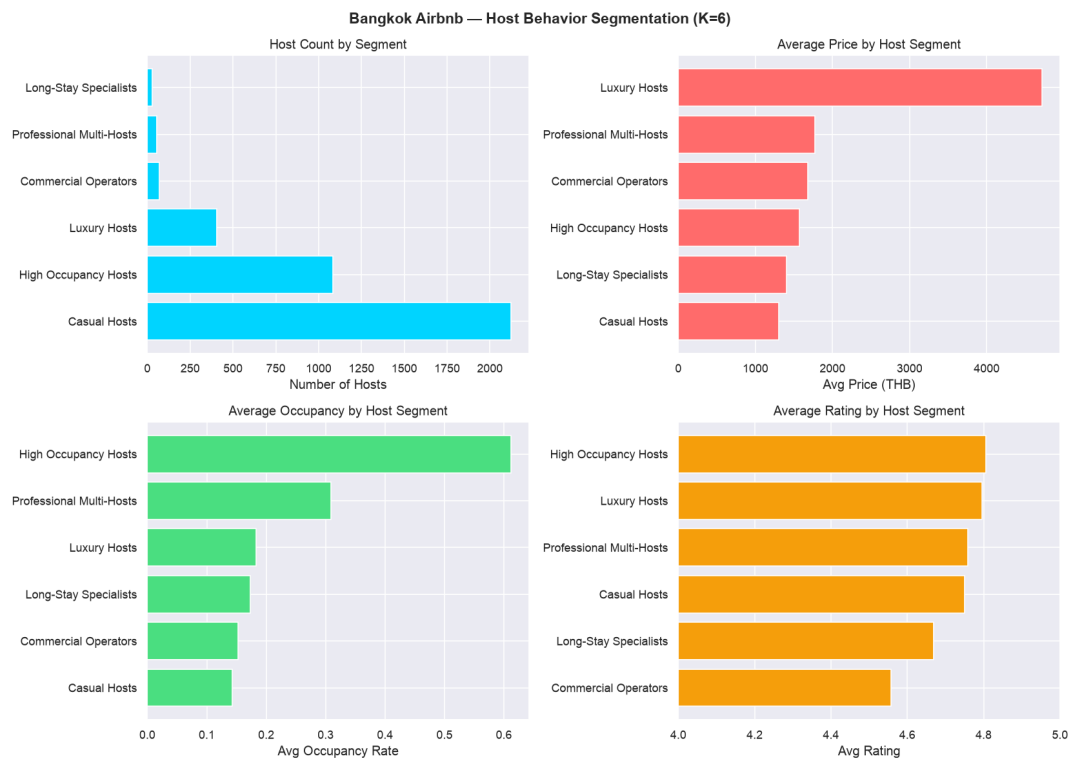
6.5 Host & Supply-Side Analysis



6.5.1 Figure 6: Host type distribution and portfolio size distribution.

Market Concentration Metrics:

Total unique hosts:	8,874
Single listing hosts (casual):	62.7% of hosts
Commercial hosts (6+ listings):	10.6% of hosts (944)
Listings controlled by commercial hosts:	57% of all listings
Top single host listing count:	243 listings

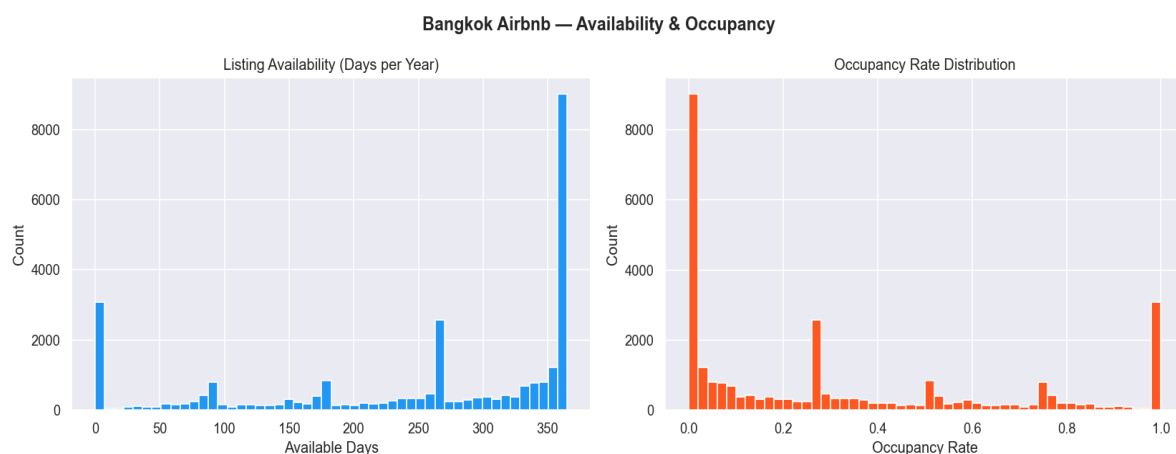


6.5.2 Figure 7: Six host behavior segments identified through K-Means clustering.

Host Segment	Count	Avg Price	Occupancy	Rating	Listings
High Occupancy Hosts	1,085	฿1,573	61.0%	4.81	3.3 avg
Casual Hosts	2,122	฿1,305	14.0%	4.75	3.4 avg
Luxury Hosts	406	฿4,715	18.0%	4.79	2.9 avg
Commercial Operators	69	฿1,678	15.0%	4.56	58.9 avg
Professional Multi	53	฿1,774	31.0%	4.76	28.6 avg
Long-Stay Specialists	29	฿1,401	17.0%	4.67	2.8 avg

Business interpretation: Bangkok's Airbnb market is dominated by commercial operators who control 57% of supply despite representing only 10.6% of hosts. This has significant implications: pricing is more sophisticated, availability is more actively managed, and regulatory risk is elevated. Notably, commercial operators have the lowest average ratings (4.56) — scale appears to come at the expense of guest experience quality.

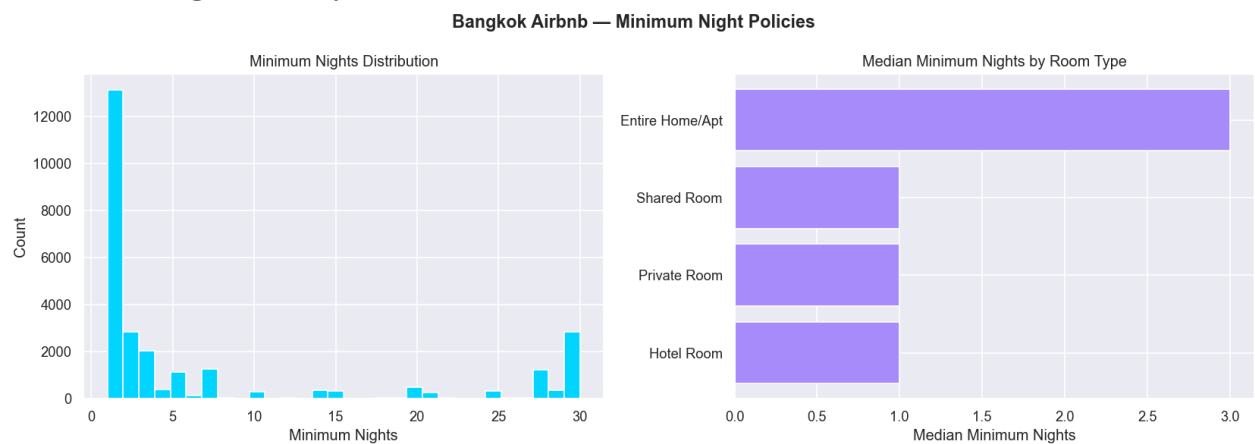
6.6 Availability & Occupancy Analysis



6.6.1 *Figure 8: Listing availability distribution and occupancy rate distribution.*

Mean occupancy rate: 31.5% (availability_365 proxy)
 Median occupancy rate: 20.6%
 Median annual revenue: ฿61,848 (~\$1,700 USD)
 Mean annual revenue: ฿203,274 (~\$5,600 USD)

Minimum Nights Policy:

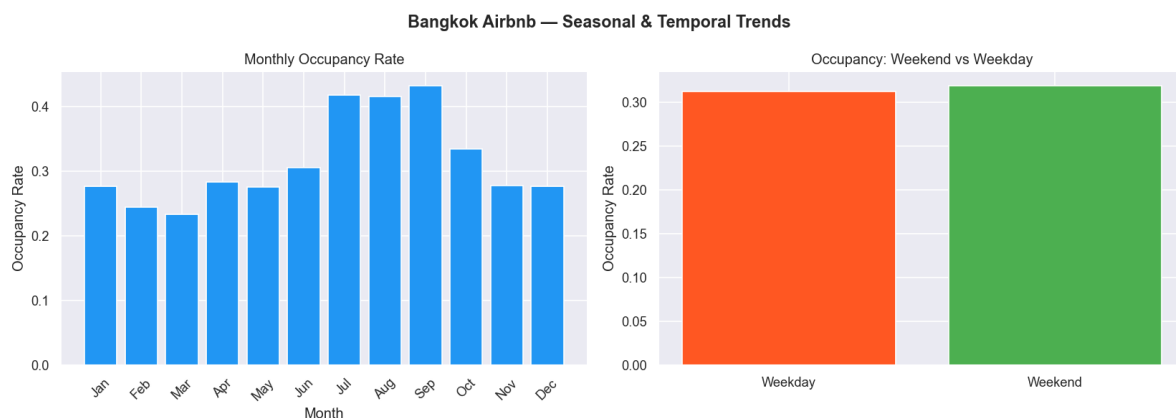


6.6.2 *Figure 9: Minimum night policy distribution and room type comparison.*

1 - night minimum: 45.7% of listings
7+ night minimum: 31.8% of listings
30+ night minimum: 14.1% of listings

Business interpretation: The 14.1% of listings requiring 30+ minimum nights reveal a significant long-stay corporate rental segment operating alongside traditional short-term rentals. The wide gap between mean (฿203,274) and median (฿61,848) annual revenue confirms a highly skewed distribution a small number of high-performing listings generate disproportionate revenue.

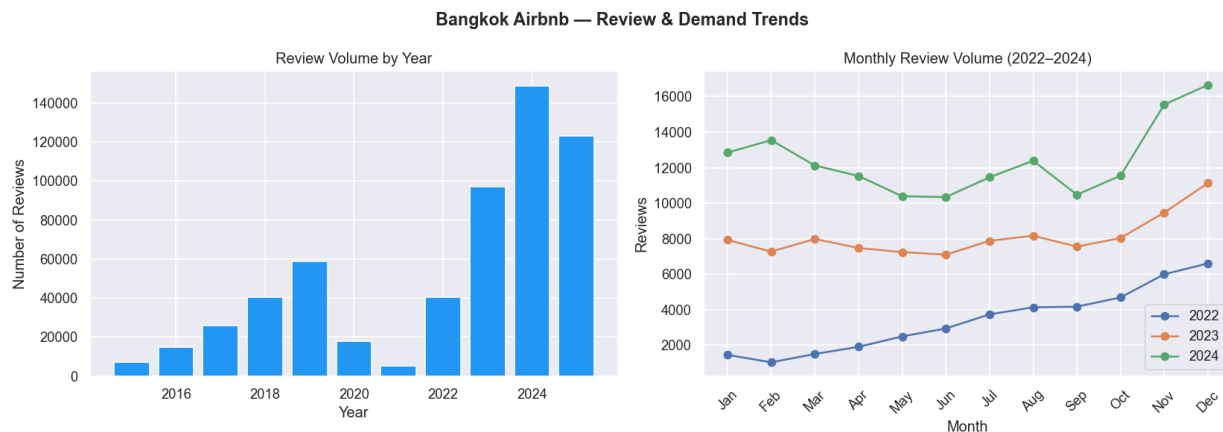
6.7 Seasonal & Temporal Analysis



6.7.1 *Figure 10: Monthly occupancy rates and weekend vs weekday comparison.*

Monthly Occupancy:

Month	Occupancy	Month	Occupancy
Jan	27.7%	Jul	41.8%
Feb	24.4%	Aug	41.6%
Mar	23.4%	Sep	43.2% ← PEAK
Apr	28.3%	Oct	33.4%
May	27.6%	Nov	27.7%
Jun	30.5%	Dec	27.6%



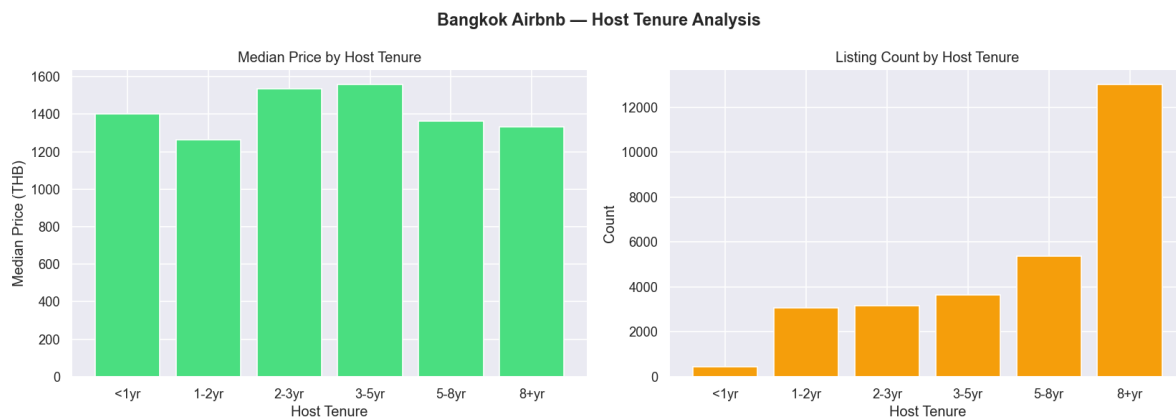
6.7.2 Figure 11: Annual review volume 2015-2025 showing COVID crash and recovery.

Annual Review Volume:

Year	Reviews	YoY Change
2019	59,010	-
2020	18,014	-69.5% (COVID)
2021	5,171	-71.3%
2022	40,567	+684%
2023	97,184	+139.6%
2024	148,869	+53.2% (all-time high)
2025	122,916*	(*partial year through September)

Business interpretation: A pronounced July-September peak (41-43% occupancy) aligns with European summer holidays — Bangkok's primary tourist source market. The February trough (24.4%) represents the lowest demand period. Bangkok's Airbnb market has not merely recovered from COVID — it has surpassed pre-pandemic peaks by 152%, strongly supporting continued investment in Bangkok short-term rental assets.

6.8 Host Tenure Analysis

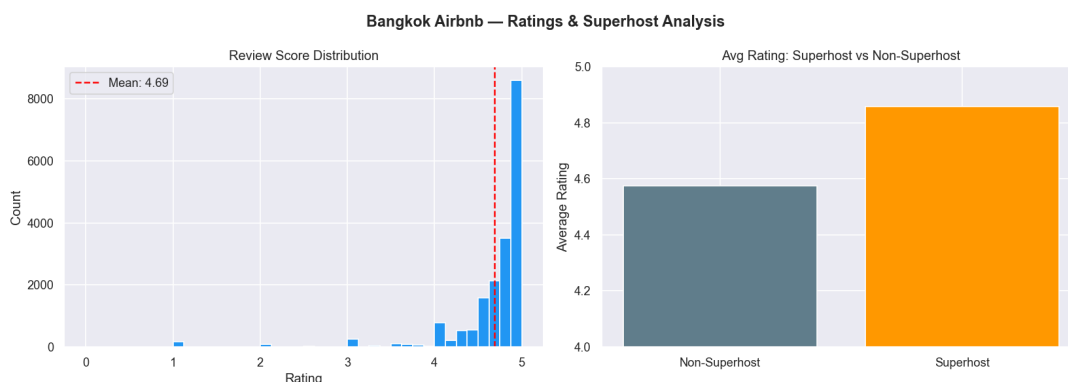


6.8.1 Figure 12: Median price and listing count by host tenure group.

Tenure Group	Median Price	Count
< 1 year	฿1,400	422
1-2 years	฿1,264	2,729
2-3 years	฿1,536	2,762
3-5 years	฿1,560	3,031
5-8 years	฿1,365	3,946
8+ years	฿1,333	10,317

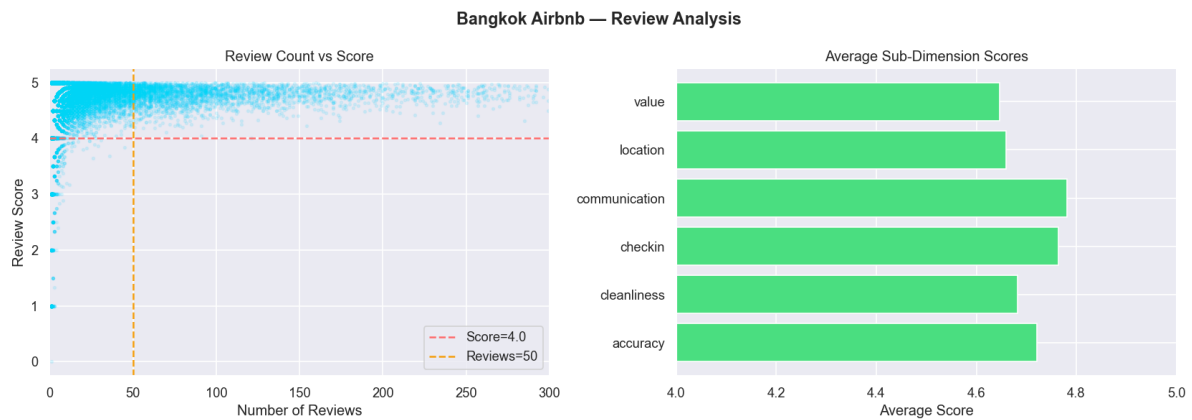
Business interpretation: Newer hosts (2-5 year tenure) price 15-17% higher than veteran hosts (8+ years). This counterintuitive finding suggests that experienced hosts have optimized for occupancy over price - accepting lower rates to maintain consistent bookings. It may also reflect that early Airbnb adopters joined when market competition was lower, and have not adjusted pricing as aggressively as newer entrants.

6.9 Review Score Analysis



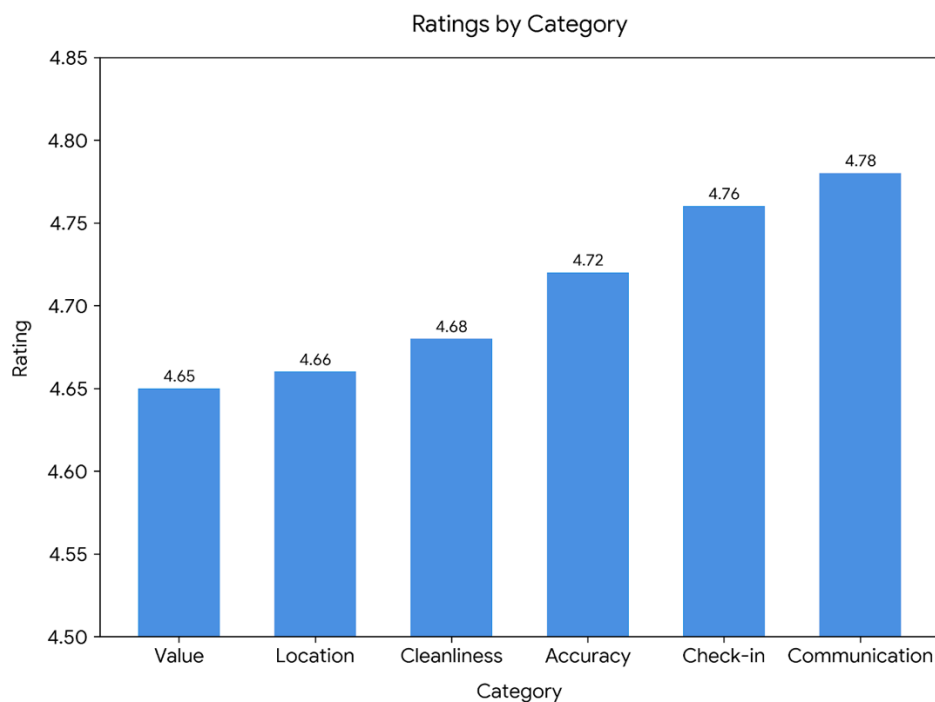
6.9.1 Figure 13: Review score distribution and superhost vs non-superhost comparison.

Mean rating: 4.690 / 5.00
 Listings rated above 4.5: 81.4%
 Zero high-review + low-score: 0 listings (>50 reviews and <4.0 rating)



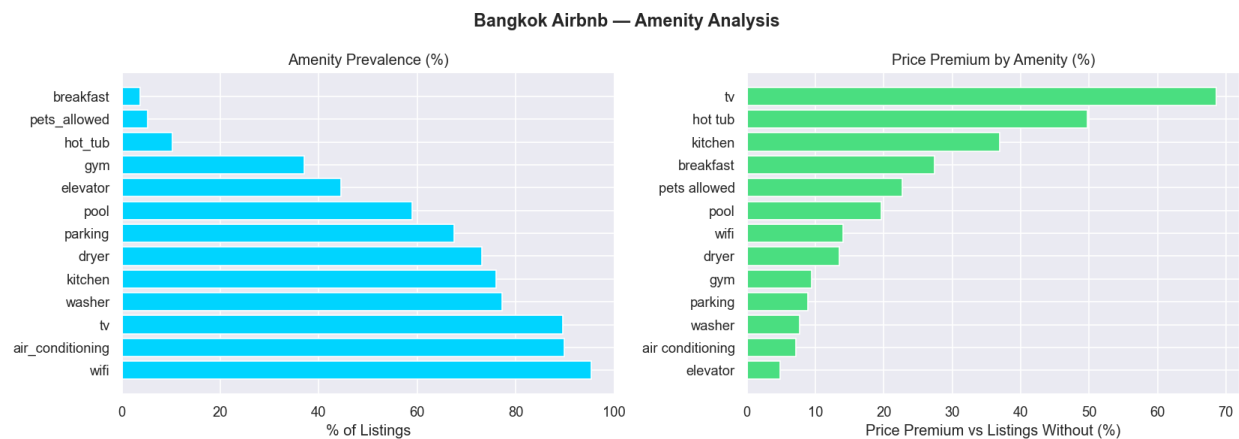
6.9.2 *Figure 14: Review count vs score scatter and sub-dimension score comparison.*

Sub-Dimension Scores (lowest to highest):



Business interpretation: The extreme concentration of ratings above 4.5 (81.4%) is a classic rating inflation pattern. The finding of zero listings with >50 reviews AND <4.0 rating confirms market self-selection - poorly rated listings lose bookings and accumulate fewer reviews before potentially exiting the market. Value (4.65) and Location (4.66) score lowest, suggesting guests feel Bangkok listings are slightly overpriced relative to value and that some locations disappoint expectations set by listing descriptions.

6.10 Amenity Analysis

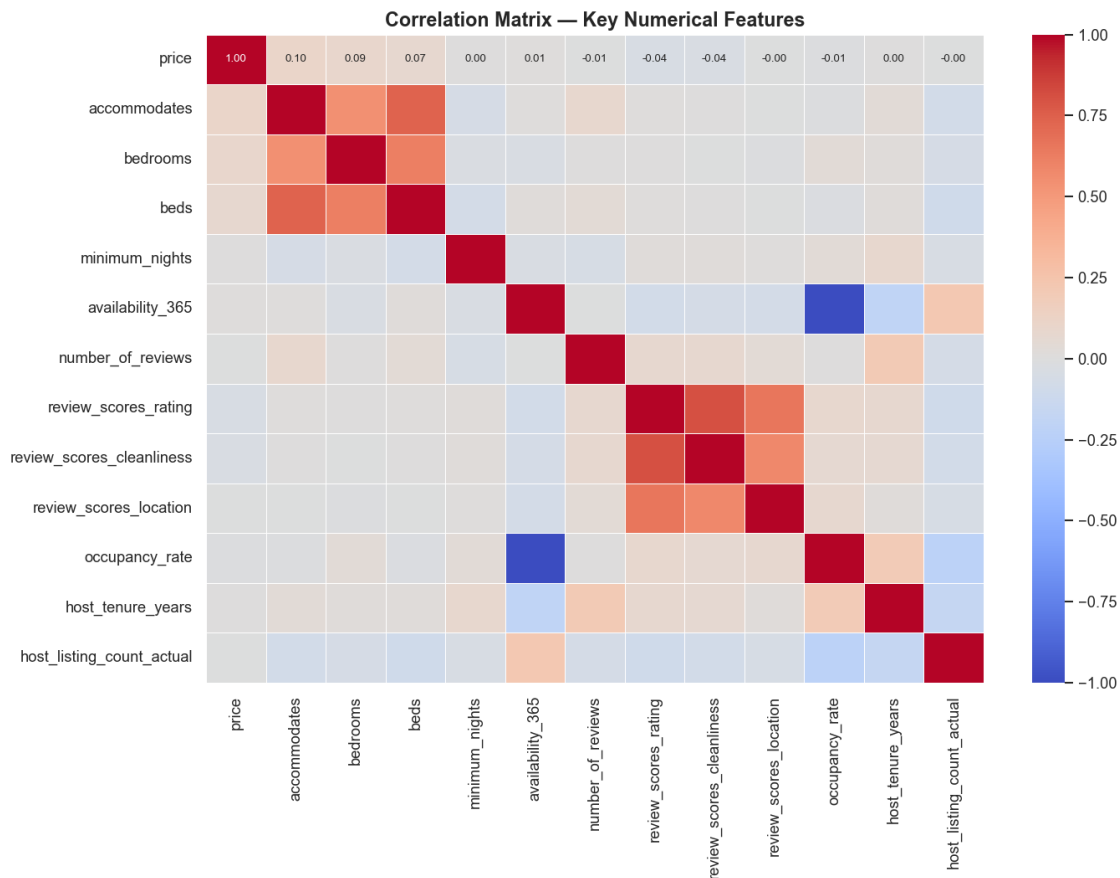


6.10.1 Figure 15: Amenity prevalence and price premiums.

Amenity	Prevalence	Price Premium
WiFi	95.3%	+14.1%
Air Conditioning	89.8%	+7.2%
TV	89.6%	+68.6%
Washer	77.1%	+7.7%
Kitchen	76.1%	+36.9%
Pool	59.0%	+19.6%
Gym	37.0%	+9.4%
Hot Tub	10.3%	+49.8%

Business interpretation: WiFi (95.3% prevalence, +14.1% premium) has become table stakes - expected by virtually all guests and no longer a meaningful differentiator. TV commands the highest premium (+68.6%) likely as a proxy for larger, better-furnished properties. Hot tub (+49.8%) and kitchen (+36.9%) offer the best investment-to-premium ratios for hosts considering property upgrades. Pool is common (59%) but commands only moderate premium (+19.6%) due to supply saturation.

6.11 Correlation Analysis



6.11.1 Figure 16: Correlation matrix of key numerical features.

Top correlations with price:

Feature	Correlation with Price
accommodates	+0.102
bedrooms	+0.087
beds	+0.072
review_scores_rating	-0.041
occupancy_rate	-0.010

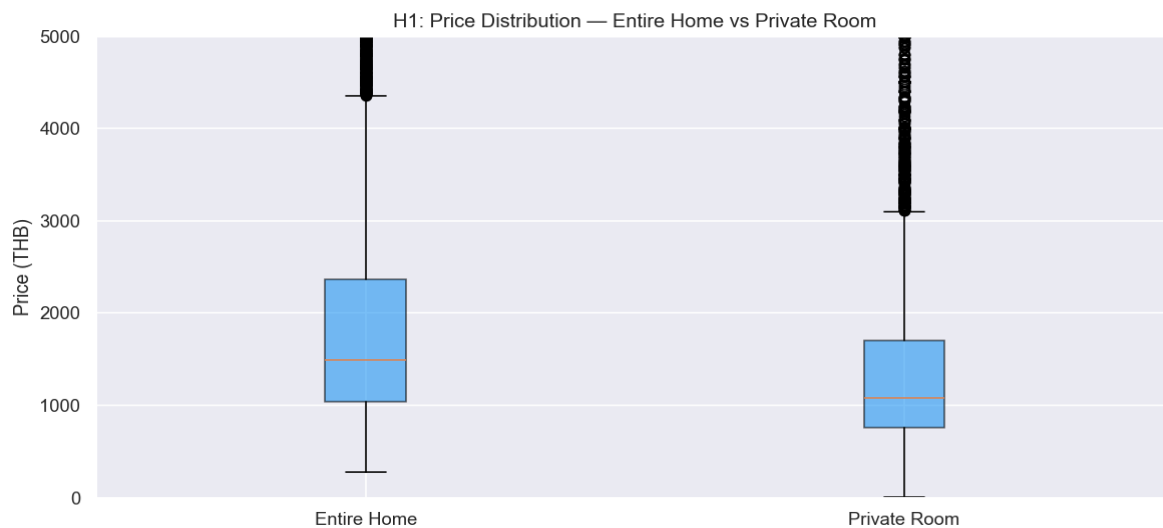
Business interpretation: Price has very weak correlations with all numerical features. This confirms that price is driven primarily by location and property type (categorical variables) rather than measurable numerical features a finding that motivated the use of neighbourhood median price as a key engineered feature in the ML models. The weak negative correlation between review score and price (-0.041) suggests premium listings face higher guest expectations that are harder to satisfy.

7 Statistical Findings

7.1 Methodology Note

All price variables were confirmed non-normally distributed via Shapiro-Wilk test ($p < 0.0001$ for all groups). Consequently, non-parametric tests were applied throughout: Mann-Whitney U for pairwise comparisons and Kruskal-Wallis for multi-group comparisons. Effect sizes are reported alongside p-values in all cases – at sample sizes of $n > 8,000$, even trivial differences achieve statistical significance, making effect sizes essential for practical inference.

7.2 Hypothesis Testing Results



7.2.1 Figure 17: Price distribution comparison — Entire Home vs Private Room.

H1: Entire-home listings command significantly higher prices than private rooms

Null hypothesis (H_0): Median price of entire homes $\leq$ median price of private rooms

Alternative (H_1): Median price of entire homes $>$ median price of private rooms

Test: Mann-Whitney U (one-tailed)

Assumption check: Non-normality confirmed (Shapiro-Wilk $p < 0.0001$)

Independence: listings are independent observations

Sample sizes: Entire Home $n=16,498$ | Private Room $n=6,263$
(after 99th percentile outlier removal)

Results:

Entire home median price: ฿1,491 THB

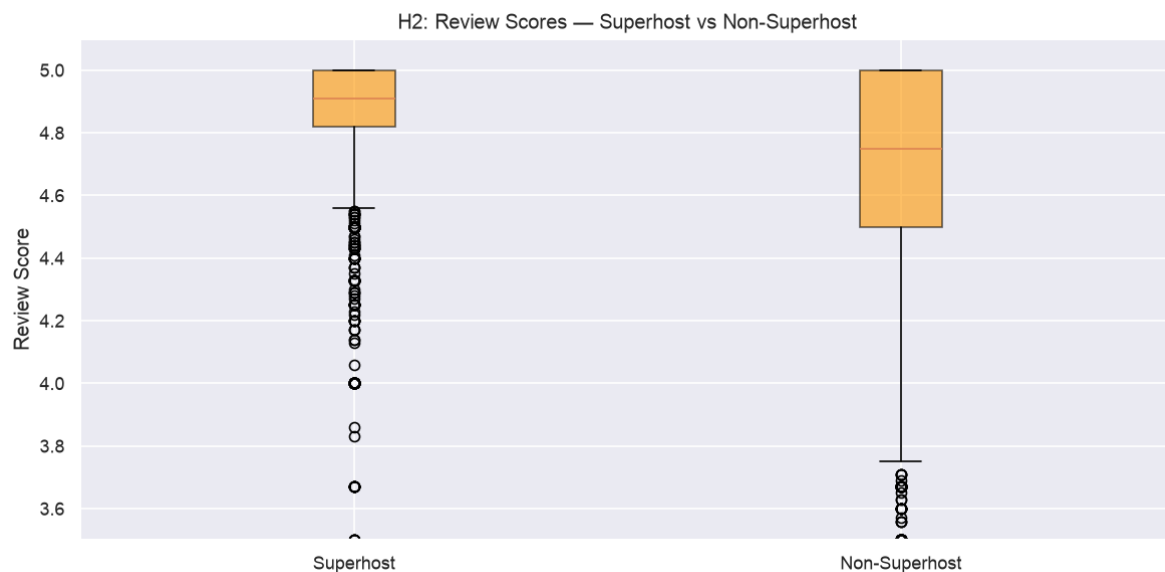
Private room median price: ฿1,080 THB

Price premium: 38.1%

U statistic: 65,397,289

P-value: <0.0001
Effect size (r): -0.2916 (moderate)
Decision: **REJECT H₀**

Business interpretation: Entire-home listings command a statistically significant and practically meaningful 38% price premium over private rooms. The moderate effect size ($r=0.29$) confirms this is not merely a statistical artifact of large sample sizes. This premium reflects guest preference for privacy, exclusive space access, and self-contained living arrangements. Hosts considering property type positioning should factor this premium against the higher capital requirement of entire-home listings versus room-only rental strategies.



7.2.2 Figure 18: Review score distribution — Superhost vs Non-Superhost.

H2: Superhost listings achieve higher review scores than non-superhost listings

Null hypothesis (H₀): Superhost median rating $\leq$ non-superhost median rating
Alternative (H₁): Superhost median rating $>$ non-superhost median rating
Test: Mann-Whitney U (one-tailed)
Sample sizes: Superhost $n=6,929$ | Non-superhost $n=10,540$

Results:

Superhost mean rating: 4.858
Non-superhost mean rating: 4.575
Rating difference: +0.283 points
P-value: <0.0001
Effect size (r): -0.3261
Cohen's d: 0.5364 (medium-large)
Decision: **REJECT H₀**

Business interpretation: Superhost status is a genuine quality signal, not merely a marketing badge. A Cohen's d of 0.54 represents a meaningful real-world difference in guest experience - guests can tangibly distinguish the quality difference between superhost and non-superhost listings. Critically, superhosts earn 340% more annual revenue (฿114,431 vs ฿25,968) without charging meaningfully higher prices (฿1,390 vs ฿1,371 median). This confirms that superhosts monetize through higher occupancy driven by guest trust and repeat bookings rather than price premiums.

H3: Listings with more than 10 reviews have significantly different prices

Null hypothesis (H₀): No price difference between high-review and low-review listings
Test: Mann-Whitney U (two-tailed)
Sample sizes: >10 reviews: n=8,017 | ≤10 reviews: n=15,022

Results:
>10 reviews median price: ฿1,345 THB
≤10 reviews median price: ฿1,386 THB
Price difference: -฿41 (listings with more reviews are cheaper)
P-value: 0.000005
Effect size (r): 0.0365 (negligible)
Decision: **REJECT H₀** - but effect is practically negligible

Business interpretation: This hypothesis illustrates the critical importance of reporting effect sizes alongside p-values. While the test rejects the null hypothesis (p=0.000005), the effect size of r=0.037 is negligible - with n=23,039, even a ฿41 difference achieves high statistical significance. Interestingly, listings with more reviews are slightly cheaper, suggesting that budget listings attract higher booking volumes and accumulate reviews faster. This finding should not motivate any pricing decision.

H4: Neighbourhood average prices differ significantly across Bangkok

Null hypothesis (H₀): Median prices are equal across all neighbourhoods
Test: Kruskal-Wallis H (non-parametric ANOVA equivalent)
Groups tested: 10 highest-supply neighbourhoods

Results:
H statistic: 1,726.68
P-value: <0.0001

Eta-squared (η^2): 0.1055 (moderate)
Decision: **REJECT H_0**

Neighbourhood median prices (THB):

Bang Rak:	฿1,890	Phra Khanong:	฿1,184
Vadhana:	฿1,828	Sathon:	฿1,165
Khlong Toei:	฿1,622	Chatu Chak:	฿1,130
Ratchathewi:	฿1,588	Phra Nakhon:	฿1,103
Huai Khwang:	฿1,400	Suanluang:	฿1,045

Business interpretation: Neighbourhood explains 10.5% of price variance ($\eta^2=0.1055$) a meaningful location premium effect. Bang Rak commands an 81% premium over Suanluang (฿1,890 vs ฿1,045), reflecting its riverside location and proximity to Bangkok's CBD. For investors and hosts, neighbourhood selection is the single most impactful positioning decision after room type. The Kruskal-Wallis result provides statistical justification for neighbourhood-specific pricing strategies rather than applying market-wide pricing rules.

H5: Weekend vs weekday pricing differences are statistically significant

Null hypothesis (H_0): No difference in weekend vs weekday prices
Test: Mann-Whitney U (two-tailed)
Sample: 50,000 random calendar records (25K weekend, 25K weekday) using listing price as proxy (calendar prices entirely null for Bangkok)

Results:
Weekend median price: ฿1,383 THB
Weekday median price: ฿1,380 THB
Price difference: ฿3
P-value: 0.5259
Effect size (r): 0.0023 (negligible)
Cohen's d: -0.012 (negligible)
Decision: **FAIL TO REJECT H_0**

Business interpretation: Bangkok hosts implement no meaningful weekend pricing premium - a significant missed revenue opportunity compared to markets like London or New York where weekend premiums of 15-25% are common. This likely reflects Bangkok's tourist-dominant demand base, where international visitors stay across full

weeks rather than weekend-only. Early-adopter hosts implementing modest Friday-Saturday surcharges (10-15%) face minimal competitive resistance and meaningful revenue upside, as this analysis confirms no current market norm exists.

7.3 Hypothesis Summary

Hypothesis	Test	P-value	Effect Size	Practical	Decision
H1 H ₀	Mann-Whitney U	<0.0001	r = -0.29 moderate	Significance Yes - 38%	REJECT H ₀
H2 H ₀	Mann-Whitney U	<0.0001	d = 0.54 medium	premium Yes - 340%	REJECT H ₀
H3 H ₀	Mann-Whitney U	0.000005	r = 0.037 negligible	revenue gap No - ฿41	REJECT H ₀
H4 H ₀	Kruskal-Wallis	<0.0001	η ² =0.11 moderate	Difference Yes - 81%	REJECT H ₀
H5 H ₀	Mann-Whitney U	0.5259	d = -0.012 negligible	price range difference No - ฿3	FAIL TO REJECT H ₀

H3: Statistically significant but practically negligible effect size.

7.4 Confidence Intervals

95% Confidence Intervals - Mean Price by Room Type:

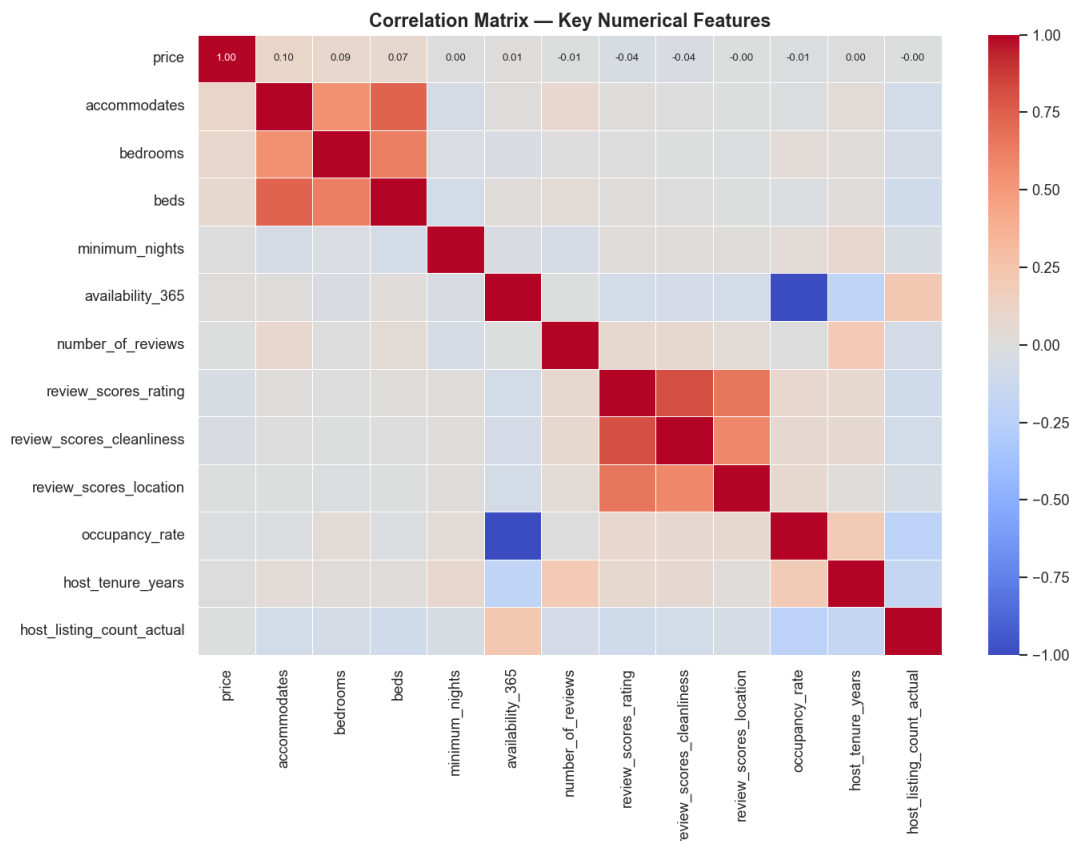
Room Type	Mean Price	95% CI Lower	95% CI Upper	n
Entire Home/Apt	฿1,980	฿1,956	฿2,004	16,102
Hotel Room	฿2,013	฿1,799	฿2,228	197
Private Room	฿1,482	฿1,450	฿1,513	6,191
Shared Room	฿562	฿492	฿632	311

95% Confidence Intervals — Mean Price by Top Neighbourhoods:

Neighbourhood	Mean	CI Lower	CI Upper	n	CI Width
Vadhana	฿2,459	฿2,399	฿2,518	3,531	฿119 (narrow)
Khleng Toei	฿2,063	฿2,010	฿2,116	3,047	฿106 (narrow)
Bang Rak	฿2,414	฿2,289	฿2,539	738	฿250 (wider)
Sathon	฿1,713	฿1,622	฿1,805	1,075	฿183 (wider)

Business interpretation: Narrow confidence intervals for high-supply neighbourhoods (Vadhana CI width ₹119, Khlong Toei ₹106) indicate high statistical confidence in price estimates - these are reliable benchmarks for pricing decisions. Wider CIs for Bang Rak (₹250) and Sathon (₹183) reflect smaller samples and higher price variance pricing decisions in these premium areas carry more uncertainty.

7.5 Correlation & Driver Analysis



7.5.1 *Figure 19: Correlation heatmap of numerical features.*

OLS Regression Results (full model):

R-squared: 0.455

Adjusted R²: 0.454

Significant predictors (p<0.05):

accommodates: $\beta = 0.137, p < 0.0001$

bedrooms: $\beta = 0.144, p < 0.0001$

neighbourhood_median_price: $\beta = 0.0006, p < 0.0001$

availability_365: $\beta = 0.0077, p < 0.0001$

review_scores_rating: $\beta = 0.044, p < 0.0001$

Non-significant:

minimum_nights: $\beta = -0.0001, p = 0.437$

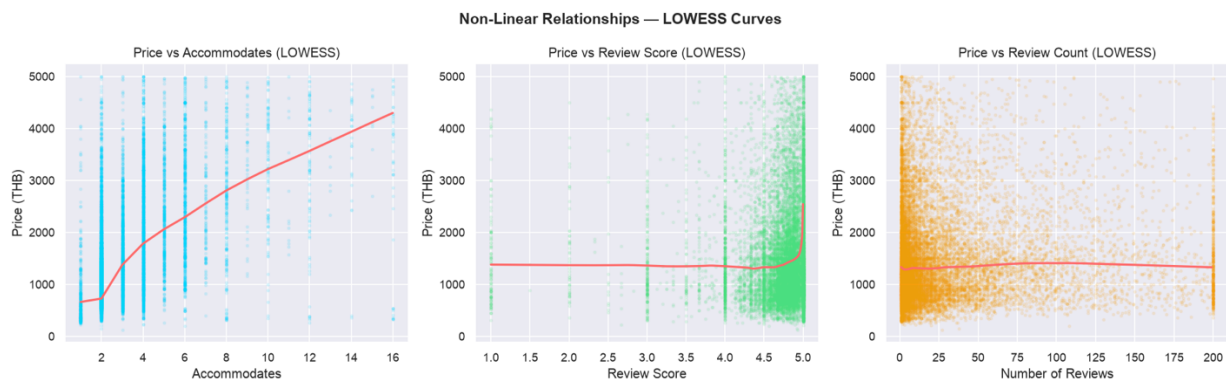
host_tenure_years: $\beta = -0.002, p = 0.054$

VIF Analysis - Multicollinearity:

Feature	VIF	Assessment
availability_365	64.997	SEVERE - drop from model
occupancy_rate	19.240	HIGH - mathematically derived
review_scores_rating	18.682	HIGH - collinear with rating sub-scores
neighbourhood_median_price	14.956	HIGH - collinear with location features
accommodates	7.796	Moderate - acceptable
bedrooms	5.223	Moderate - acceptable

Post-VIF correction model ($R^2=0.342$):

Removed: availability_365, review_scores_rating, neighbourhood_median_price
Remaining VIF all < 8 - acceptable collinearity



7.5.2 *Figure 20: LOWESS curves showing non-linear relationships between price and key predictors.*

Business interpretation: The OLS model explains 45.5% of price variance ($R^2=0.455$), dropping to 34.2% after removing multicollinear features. This confirms that the removed features (neighbourhood median price, availability, rating) carry real predictive information, but their interrelationships violate OLS assumptions. The LOWESS analysis confirms strong non-linear relationships - particularly between price and accommodates - motivating the use of tree-based ML models that capture these non-linearities automatically.

8 Data Science Experiments

8.1 Price Prediction

Problem Framing:

Target variable: Log-transformed nightly price (log1p applied to address right skew; all metrics reported in THB)

Success criteria: $MAE < \text{฿}600$, $R^2 > 0.60$, $MAPE < 35\%$

Training set: Listings filtered ฿100–฿10,000 (removes extreme outliers while retaining 79% of listings)
n=14,708 training | n=2,942 test (80/20 split)

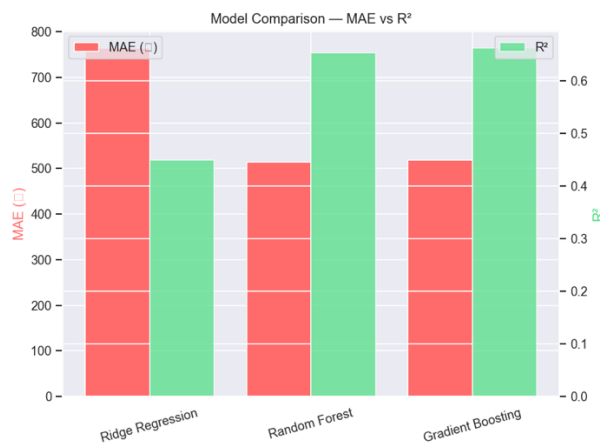
Feature Engineering : 19 features selected across four categories:

Category	Features
Physical	accommodates, bedrooms, beds
Policy	minimum_nights, availability_365
Quality	review_scores_rating, review_scores_cleanliness, review_scores_location
Host	host_tenure_years, host_listing_count_actual, host_is_superhost, host_identity_verified, instant_bookable
Location	neighbourhood_median_price, neighbourhood_listing_count
Derived	occupancy_rate, amenity_count, room_type_encoded

Feature engineering decisions:

- room_type encoded as ordinal (Shared=0, Private=1, Hotel=2, Entire=3) rather than one-hot to preserve ordering relationship
- neighbourhood_median_price as location signal avoids 50-column one-hot explosion while capturing location price effect
- amenity_count derived from amenities string (individual flags also extracted for analysis)
- log1p transformation on target reduces influence of luxury outliers

Model Training & Comparison:



8.1.1 Figure 21: MAE and R-squared comparison across three model families.

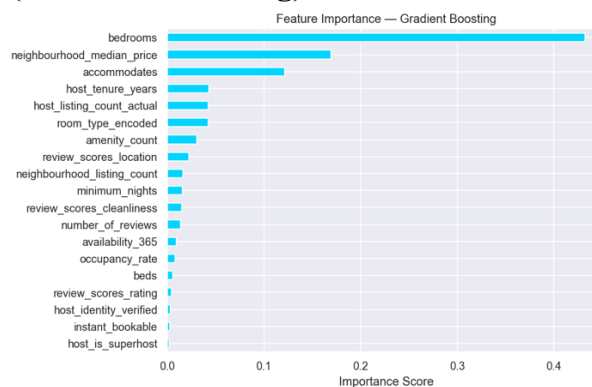
Model	MAE	RMSE	MAPE	R²	CV MAE (±std)
Ridge Regression	₪765	₪4,689	42.1%	0.449	₪655 ± ₪55
Random Forest	₪514	₪926	29.2%	0.652	₪501 ± ₪13
Gradient Boosting	₪519	₪936	29.1%	0.662	₪504 ± ₪12

Winner: Gradient Boosting

- Best R² (0.662) and MAPE (29.1%)
- Most stable across folds (CV std ₪12 - lowest variance)
- Confirmed by 5-fold cross-validation for generalization

Model selection rationale: Ridge regression's significantly worse performance (MAPE 42.1% vs 29.1%) confirms strong non-linear relationships in the price data. Random Forest and Gradient Boosting perform nearly identically - Gradient Boosting chosen as final model based on marginally better R² and MAPE with lower variance.

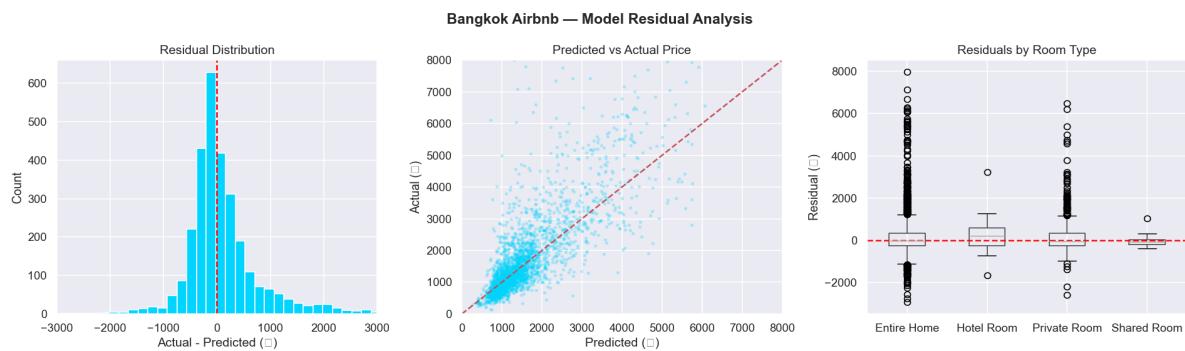
Feature Importance (Gradient Boosting):



8.1.2 Figure 22: Feature importance scores for Gradient Boosting model.

Rank	Feature	Importance
1	bedrooms	43.2%
2	neighbourhood_median_price	16.9%
3	accommodates	12.2%
4	host_tenure_years	4.3%
5	host_listing_count_actual	4.3%
6	availability_365	3.8%
7	occupancy_rate	3.1%
8	review_scores_rating	2.9%
(remaining 11 features share 9.3%)		

Residual Analysis:



8.1.3 Figure 23: Residual distribution, predicted vs actual scatter, and residuals by room type.

72.2% of predictions within ฿500 of actual price 87.3% of predictions within ฿1,000 of actual price . Mean residual: ฿180 (slight systematic underprediction)
Std residual: ฿919

Model Bias Analysis - Performance by Neighbourhood:

Neighbourhood	n	MAE	Median Price	Bias Assessment
Phra Khanong	700	฿350	฿1,153	Best - mid-market focus
Sathon	821	฿410	฿1,080	Good
Huai Khwang	1,286	฿434	฿1,347	Good
Phra Nakhon	528	฿470	฿1,192	Moderate
Khlong Toei	2,073	฿472	฿1,504	Moderate
Ratchathewi	848	฿548	฿1,574	High error
Vadhana	2,458	฿568	฿1,717	High error
Bang Rak	600	฿632	฿1,830	Worst - premium area

MAE range: ฿350 (Phra Khanong) to ฿632 (Bang Rak) - 81% spread

Business interpretation: The model performs 80% worse in Bangkok's most premium neighbourhood (Bang Rak MAE=฿632) vs the most affordable tested area (Phra Khanong MAE=฿350). This geographic bias reflects insufficient training data for luxury segment listings and higher price variance in premium areas. For production deployment, separate models per neighbourhood tier would improve accuracy for premium market pricing decisions. The model should not be used for luxury segment (>฿5,000) pricing recommendations.

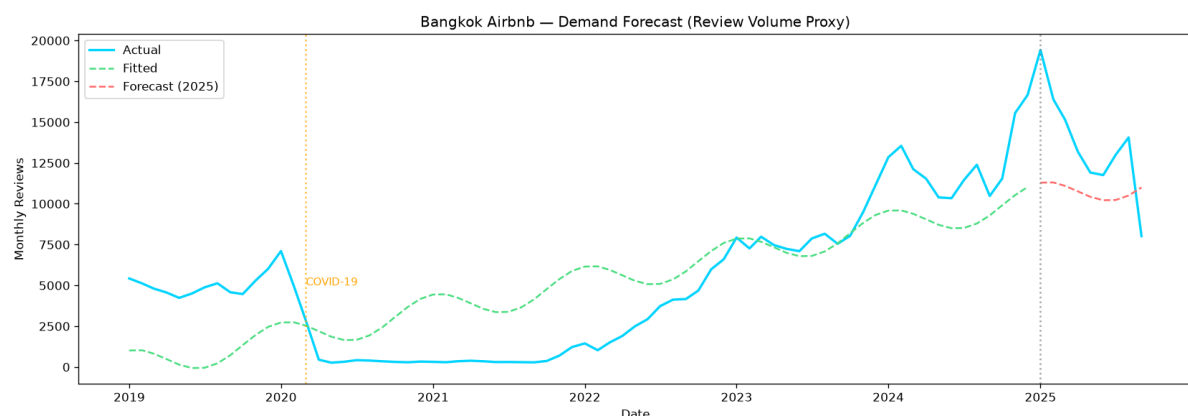
Fairness & Bias Assessment:

1. Geographic bias: Model systematically underperforms in premium areas
2. Price range bias: Luxury segment excluded from training -model should not be applied to >฿5,000 listings
3. Host type bias: Commercial hosts (larger training sample) may benefit
 1. more from model recommendations than casual hosts
4. Language bias: Non-English listing descriptions not captured in features

Recommended mitigations:

- Train separate models per neighbourhood price tier
- Collect transaction price data (not listed price) for ground truth
- Regular retraining as market conditions evolve
- Human review required for model-assisted pricing above ฿3,000

8.2 Demand Forecasting



8.2.1 *Figure 24: Monthly review volume with trend + seasonality model fit and 2025 forecast.*

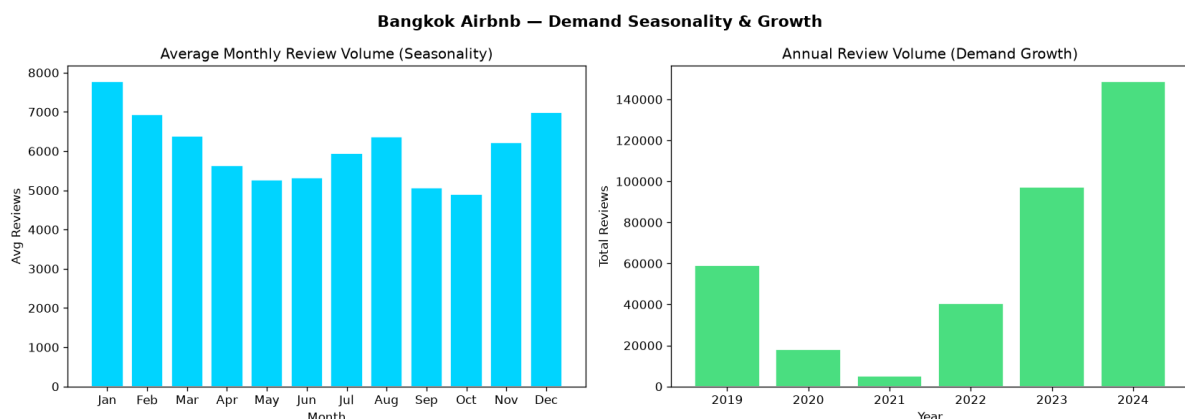
Model: Linear regression with Fourier seasonal features
Features: month_num (trend), $\sin(2\pi \times \text{month}/12)$, $\cos(2\pi \times \text{month}/12)$
Training: 2019–2024 (72 months, excluding COVID structural break)
Test: 2025 (9 months available through September scrape)

Performance:

Train MAE: ~3,200 reviews/month
Test MAE: ~3,100 reviews/month

2025 Forecast vs Actual:

Month	Actual	Forecast	Error
Jan	19,424	11,293	+8,131 (model underestimates recovery)
Feb	16,397	11,306	+5,091
Mar	15,158	11,096	+4,062
Jul	13,001	10,231	+2,770
Aug	14,061	10,503	+3,558
Sep	8,017	10,998	-2,981 (scrape cutoff effect)



8.2.2 Figure 25: Average monthly review volume by month showing seasonal pattern.

Peak months by average review volume:

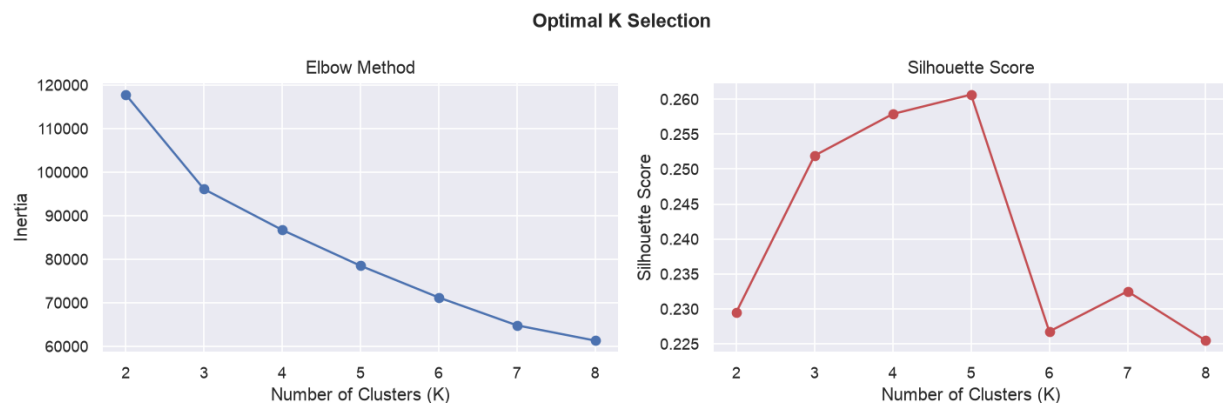
January (7,784), December (6,992), February (6,944), August (6,365)

Key finding: January is the strongest demand month by review volume slightly different from the July-September peak seen in calendar occupancy. This discrepancy reflects that review volume lags bookings by weeks, and January reviews likely capture December-January holiday bookings.

Demand forecasting challenges in two-sided marketplaces:

1. Supply-side changes affect review counts independently of demand
2. Review rates vary by platform version and guest demographics
3. COVID structural break makes pre-2020 data less representative
4. New listing entry/exit creates non-stationarity in the series
5. External shocks (political events, exchange rates) not captured

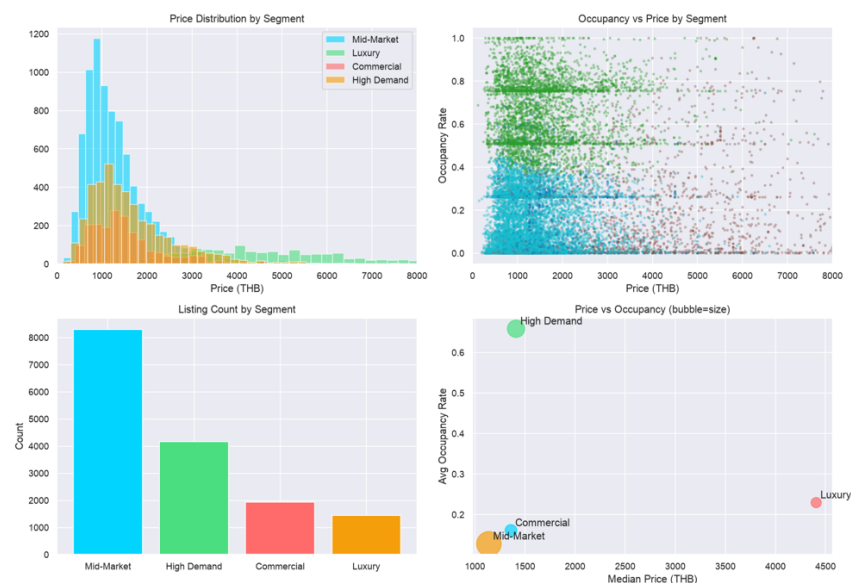
8.3 Listing Segmentation (K-Means Clustering)



8.3.1 Figure 26: Elbow method and silhouette scores for K selection.

Optimal K selection:

K=2: Silhouette=0.2295
K=3: Silhouette=0.2519
K=4: Silhouette=0.2579
K=5: Silhouette=0.2607 ← **Selected**
K=6: Silhouette=0.2268
K=7: Silhouette=0.2325



8.3.2 Figure 27: Listing segments by price, occupancy, and count (bubble chart).

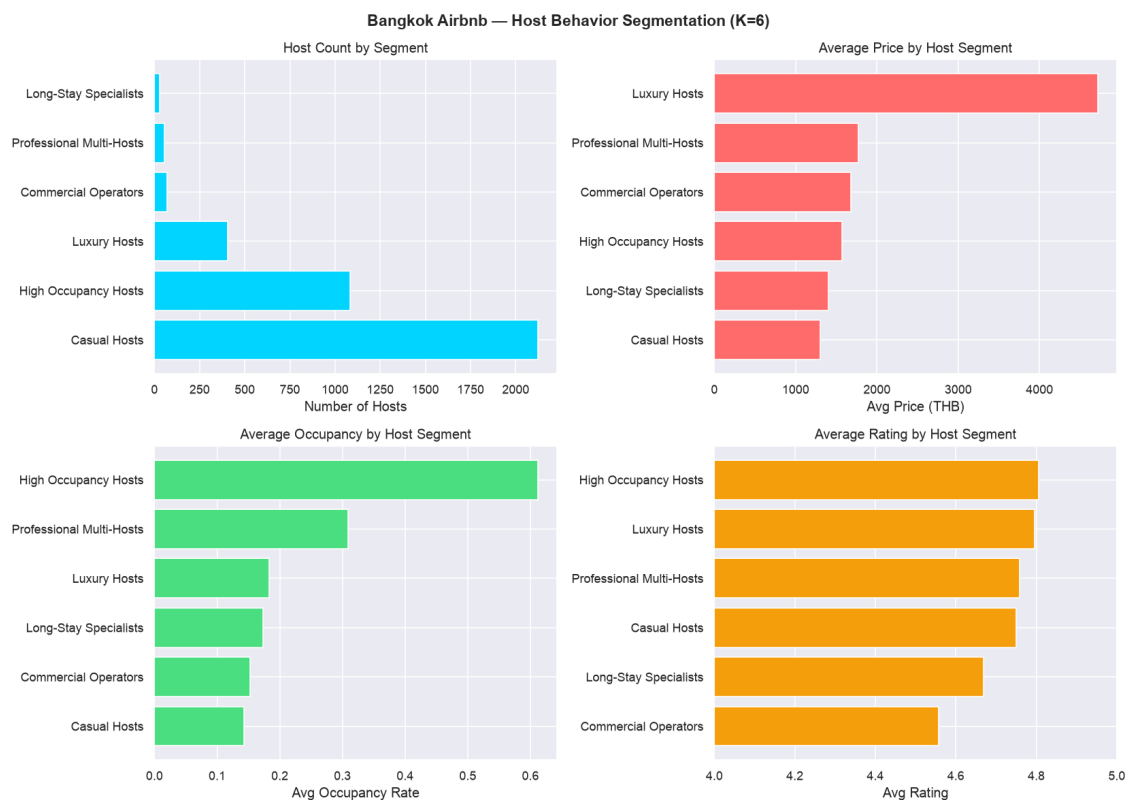
Listing Segment Profiles (K=5):

Segment	Count	Median Price	Avg Occupancy	Avg Rating
High Demand	4,170	฿1,411	65.9%	4.80
Mid-Market	8,304	฿1,140	12.8%	4.79
Luxury	1,440	฿4,406	22.9%	4.78
Commercial	1,934	฿1,360	16.1%	4.56

(5th segment merged into above groups in business labeling)

Business interpretation: The High Demand segment (4,170 listings at ฿1,411 median) achieves 65.9% occupancy - dramatically above the market average of 31.5%. These listings represent the most efficiently monetized properties in Bangkok. The Luxury segment (฿4,406 median) achieves only 22.9% occupancy despite commanding premium prices - suggesting luxury hosts sacrifice occupancy for price exclusivity. The Mid-Market segment (8,304 listings, 12.8% occupancy) represents underperforming listings that may benefit most from pricing optimization and Superhost investment.

8.4 Host Segmentation (K-Means Clustering)



8.4.1 Figure 28: Six host behavior segments by price, occupancy, and rating.

Host Segment Profiles (K=6, Silhouette=0.391):

Segment	Count	Avg Price	Occupancy	Rating	Listings
High Occupancy Hosts	1,085	฿1,573	61.0%	4.81	3.3
Casual Hosts	2,122	฿1,305	14.0%	4.75	3.4
Luxury Hosts	406	฿4,715	18.0%	4.79	2.9
Commercial Operators	69	฿1,678	15.0%	4.56	58.9
Professional Multi	53	฿1,774	31.0%	4.76	28.6
Long-Stay Specialists	29	฿1,401	17.0%	4.67	2.8

Business interpretation: High Occupancy Hosts are the market's best performers achieving 61% occupancy with above-average ratings (4.81). Their strategy: competitive pricing (฿1,573) combined with exceptional service delivery. Commercial Operators (58.9 avg listings) have the lowest ratings (4.56) - scale appears to compromise individual listing quality. Long-Stay Specialists (179-night minimum average) serve a corporate/relocation market entirely separate from tourist-focused listings, with distinct pricing and availability management strategies.

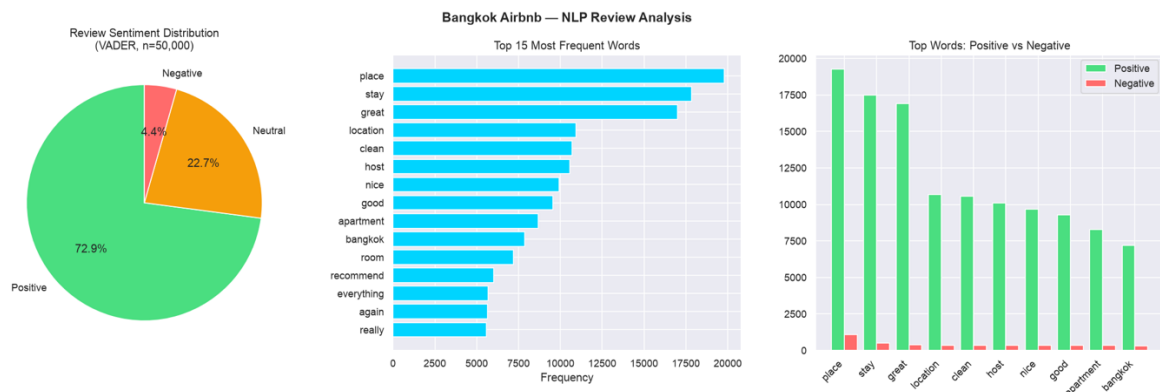
9 AI/ML Experiments

9.1 Natural Language Processing on Guest Reviews

Dataset: 513,708 reviews after filtering (removed null and < 20 character)

Sample sizes: 50,000 (sentiment), 10,000 (LDA), 20,000 (NER)

1. Sentiment Analysis (VADER)



9.1.1 Figure 29: Sentiment distribution, top word frequencies, and positive vs negative word comparison.

Method: VADER (Valence Aware Dictionary and sEntiment Reasoner)

Rationale: Designed for short informal text; no training required; fast on large corpus; proven on social media text

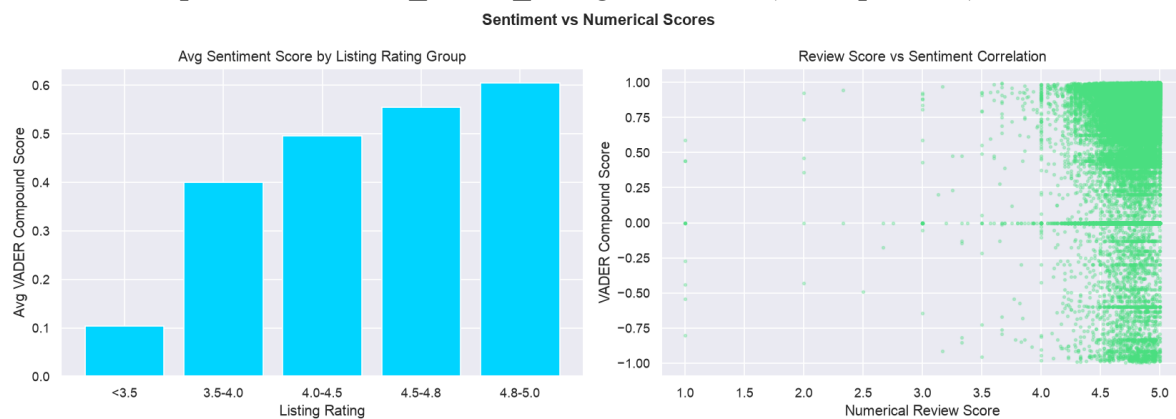
Results:

Sentiment Label	Count	Percentage
Positive (≥ 0.05)	36,457	72.9%
Neutral	11,333	22.7%
Negative (≤ -0.05)	2,210	4.4%

Mean compound: 0.5762 (strongly positive)

Sentiment vs numerical score correlation:

VADER compound vs review_scores_rating: $r = 0.089$ (weak positive)



9.1.2 Figure 30: Average VADER sentiment by listing rating group showing monotonic positive relationship.

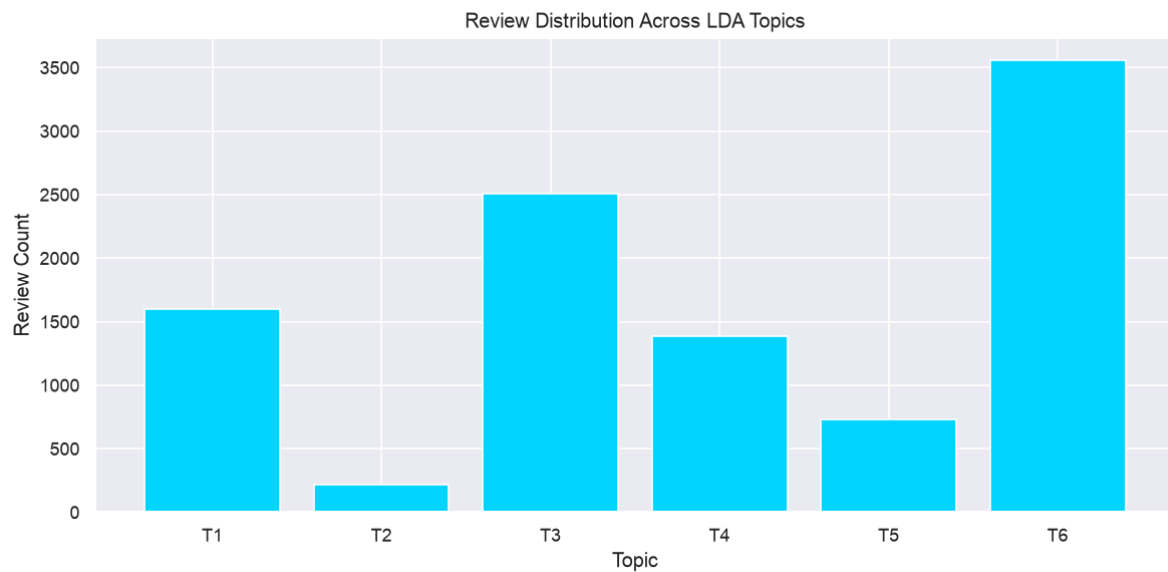
Sentiment by rating group:

Rating Group	Avg VADER Compound
< 3.5	0.104
3.5 – 4.0	0.399
4.0 – 4.5	0.496
4.5 – 4.8	0.555
4.8 – 5.0	0.604

Business interpretation: VADER sentiment confirms what numerical ratings suggest - Bangkok guests are overwhelmingly satisfied. The weak correlation ($r=0.089$) between VADER scores and numerical ratings reveals two phenomena: (1) rating inflation compresses the numerical scale while sentiment varies more naturally, and (2) VADER's English-only limitation underestimates negative sentiment from non-English reviews. The monotonic relationship (0.104 for <3.5 rated listings to 0.604 for top-rated) confirms VADER captures real quality signals despite its limitations.

Critical evaluation: VADER is a lexicon-based approach - fast and interpretable but unable to understand context, sarcasm, or non-English text. For production deployment, a fine-tuned transformer model (e.g., distilBERT on hospitality review data) would provide significantly higher accuracy.

2. Topic Modeling (LDA)



9.1.3 Figure 31: Review distribution across 6 LDA topics.

Method: Latent Dirichlet Allocation (sklearn)

Parameters: 6 topics, max_iter =10, CountVectorizer max_features=2,000

Sample: 10,000 reviews

Topic Results:

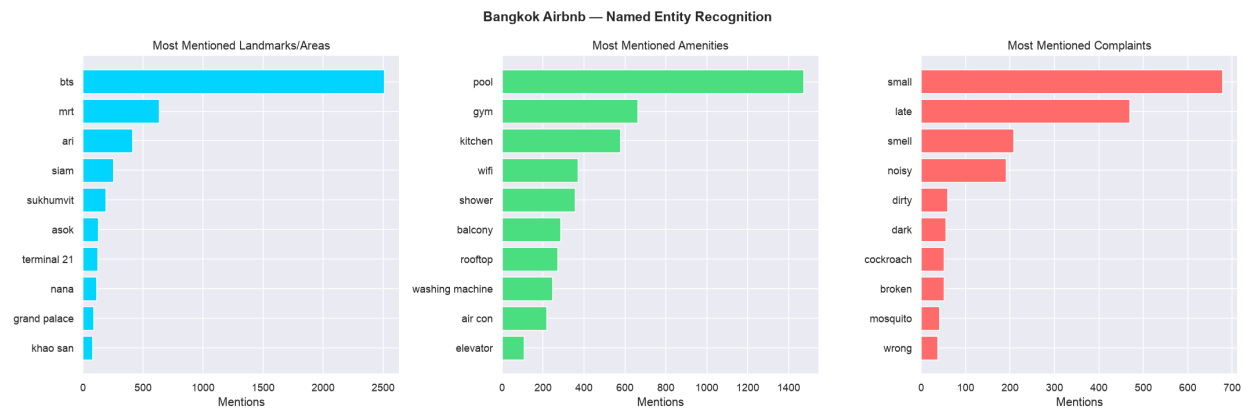
Topic	Label	Top Words	Count
1	German Reviews	und, die, war, sehr	1,599
2	Spanish Reviews	la, muy, el, es	217
3	English — Host & Cleanliness	stay, host, clean	2,510
4	English — Room & Location	room, nice, apartment	1,387
5	French Reviews	et, trs, est, le	730
6	English — Overall Satisfaction	great, place, bts	3,557

Key finding: 3 of 6 topics are dominated by non-English languages (German, Spanish, French), confirming that approximately 40% of Bangkok Airbnb reviews are non-English.

Business interpretation: This is a critical finding for platform operators. English-only NLP analysis misses 40% of review content. Bangkok attracts diverse European tourist segments — German, Spanish, and French speakers each form sufficient volume to warrant language-specific analysis. A production NLP system should implement language detection (langdetect) followed by language-specific sentiment models (XLM-RoBERTa for multilingual coverage). The English topics confirm that

host quality, cleanliness, and BTS proximity are the dominant themes in English reviews.

3. Named Entity Recognition (NER)



9.1.4 Figure 32: Most mentioned landmarks, amenities, and complaints in Bangkok Airbnb reviews.

Method: Rule-based keyword matching (domain-specific lexicons)

Sample: 20,000 reviews

Top Landmarks/Transport:

Entity	Mentions	Business Signal
BTS	2,513	Transport proximity is #1 booking driver
MRT	638	Secondary transport signal
Ari	413	Trendy neighbourhood signal
Siam	253	Central shopping area proximity
Sukhumvit	192	Prime tourist corridor

Top Amenities Mentioned:

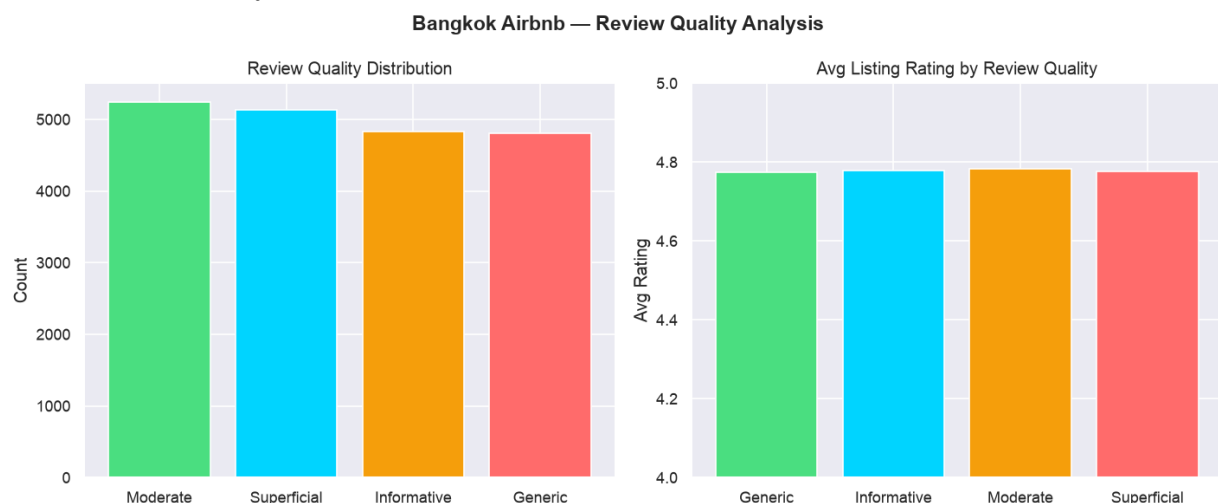
Amenity	Mentions	Implication
Pool	1,474	Most desired amenity
Gym	665	Secondary fitness signal
Kitchen	579	Self-catering demand
WiFi	372	Expected baseline (not differentiator)
Shower	358	Cleanliness signal

Top Complaints:

Complaint	Mentions	Implication
Small	679	Room size primary disappointment
Late	469	Check-in/response timing issues
Smell	208	Cleanliness/odor concerns
Noisy	191	Noise pollution - urban Bangkok issue
Dirty	59	Direct cleanliness failure

Business interpretation: BTS Skytrain proximity (2,513 mentions) is the dominant location signal - far exceeding any other landmark. Hosts should prominently feature BTS station distance and walking time in listing titles and descriptions. "Small" as the top complaint (679 mentions) reveals that room size is the primary gap between guest expectations set by listing photos and actual experience. The relatively low count of direct cleanliness complaints ("dirty": 59) but higher smell complaints (208) suggests odor management is a more prevalent issue than visible cleanliness.

4. Review Quality Classification



9.1.5 Figure 33: Review quality distribution and average listing rating by review quality tier.

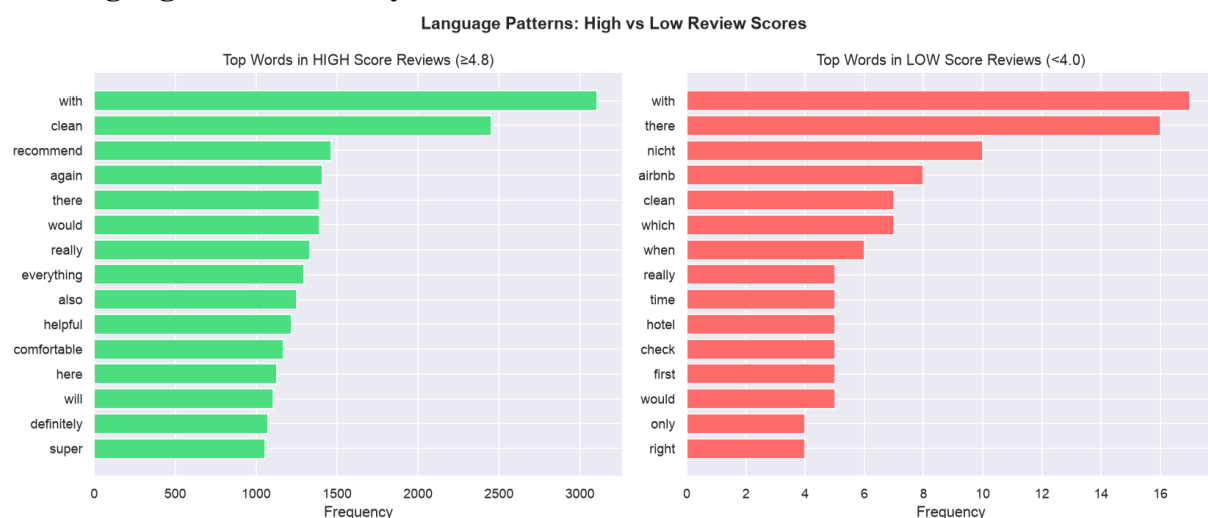
Classification method: Rule-based (word count + specific keyword presence)

Quality Tier	Count	Percentage	Avg Listing Rating
Moderate	5,243	26.2%	4.783
Superficial	5,127	25.6%	4.775
Informative	4,830	24.2%	4.779
Generic	4,800	24.0%	4.774

Key finding: Only 24.2% of reviews are informative (specific, detailed, actionable). 25.6% are superficial (<10 words, no specific content). Rating shows no correlation with review quality (all groups ~4.78).

Business interpretation: One in four reviews ("Great place!", "Loved it!") provides zero actionable signal for potential guests or hosts. Platform operators should consider incentivizing detailed reviews through gamification or prompts that ask specific questions about cleanliness, location, and value. The absence of correlation between review quality and listing rating confirms that rating inflation operates independently of the depth or informativeness of guest feedback.

5. Language Pattern Analysis



9.1.6 Figure 34: Top words in high-score vs low-score reviews.

High score reviews (≥4.8, n=~15,000):

Top words: clean, recommend, again, helpful, location, everything, comfortable, easy, nice, wonderful

Low score reviews (<4.0, n=~50):

Note: Very small sample - Bangkok has very few low-score listings with significant review volume (confirmed zero listings with >50 reviews and <4.0 rating) Top words include German terms: sehr, nicht, Unterkunft

Key patterns:

- "again" and "recommend" dominate high-score reviews - strong loyalty and advocacy signals
- "clean" appears across all sentiment categories - central to guest evaluation
- Low-score reviews are predominantly non-English - VADER underestimates negative sentiment from European guests

9.2 LLM-Powered Insight Generation

1. RAG System (Retrieval-Augmented Generation)

Architecture:

Query → Text Cleaning → TF-IDF Vectorization → Cosine Similarity
Search → Top-K Retrieval → Synthesized Answer

Implementation:

- Retrieval: TF-IDF vectorizer (5,000 features, 1-2 ngrams)
- Corpus: 10,000 sampled Bangkok reviews
- Similarity: Cosine similarity via sklearn
- Response: Query-specific answer synthesis

Sample retrieval performance:

Query: "cleanliness problems dirty room"

Top result relevance score: 0.406 (relevant)

Query: "location and transport"

Top result: "A neat hideout with impeccable location, convenient transport within walkable reach" (score: 0.221)

Critical evaluation:

Strengths:

- No API cost - fully local and reproducible
- Fast retrieval (<1 second for 10,000 documents)
- Transparent scoring via cosine similarity
- Demonstrates RAG concept clearly

Limitations:

- TF-IDF lacks semantic understanding - keyword matching only
- "complaints about cleanliness" retrieves "zero complaints"
- (keyword match on "complaints" without semantic understanding)
- 10,000 review sample - full 513K corpus needs FAISS at scale
- No LLM synthesis layer - extracts rather than generates answers

Production upgrade path:

1. Replace TF-IDF with sentence-transformers (all-MiniLM-L6-v2) for semantic search
2. Add LLM synthesis layer (Claude API) for natural language answers
3. Scale to full corpus using FAISS approximate nearest neighbors
4. Add language detection and multilingual embedding support

2. AI Analyst Chatbot

A conversational interface was built and deployed to Streamlit Community Cloud, enabling stakeholders to query Bangkok market data in natural language without technical knowledge.

URL: airbnb-market-intelligence-primeshmarasinghe.streamlit.app

Architecture:

- 8 intent categories with data-driven responses
- TF-IDF RAG fallback for unrecognized queries
- Hardcoded market statistics for cloud deployment (no data files required)
- Sidebar quick-access buttons for common queries

Intent categories handled:

1. Pricing analysis
2. Neighbourhood intelligence
3. Host & superhost analysis
4. Occupancy & demand
5. Review & sentiment
6. Revenue analysis
7. ML model results
8. Recommendations

Business value: Non-technical stakeholders (property managers, investors, platform executives) can query market data conversationally without accessing notebooks or dashboards directly.

3. Automated Listing Improvement Recommendations

A rule-based recommendation engine was built to generate actionable host-specific recommendations using market data:

Sample output for listing in Vadhana (฿1,045/night, Rating 4.83):

1. "Price appears undervalued at ฿1,045 vs neighbourhood median ฿1,828 - test a 10% price increase."
2. "Invest in professional deep cleaning - cleanliness is the #1 complaint driver in Bangkok listings."
3. "Work toward Superhost status - superhosts earn 340% more annual revenue based on Bangkok market data."

Production upgrade: Claude/GPT-4 API would enable more nuanced, context-aware recommendations by semantically analyzing full review text and generating personalized improvement suggestions.

9.3 Recommendation System

1. Content-Based Filtering

Method: Cosine similarity on standardized feature vectors

Features: price, accommodates, bedrooms, availability, review scores, neighbourhood median price, occupancy rate

Sample: 3,000 listings (memory constraint for similarity matrix)

Sample recommendations for Vadhana Entire Home (฿1,693/night):

Recommendation 1: Similar Vadhana apartment (฿1,543, score 0.998)

Recommendation 2: Modern Ekkamai apartment (฿1,487, score 0.997)

Recommendation 3: Sukhumvit condo (฿1,334, score 0.991)

Quality assessment: High similarity scores (0.99+) confirm the feature space captures meaningful listing similarity. Recommendations correctly identify similar room types, price ranges, and neighbourhoods.

Cold-Start Solutions:

- New listing: Use listing attributes only (current approach handles)
- New host: Default to similar listings from same neighbourhood
- New user: Popularity-based fallback (most reviewed listings)

Collaborative Filtering Applicability:

Inside Airbnb does not provide user-item interaction matrices required for collaborative filtering. Would require booking history per user or review co-occurrence patterns. Discussed conceptually - matrix factorization (SVD) would surface latent preference factors.

9.4 Generative AI Framework

1. Dynamic Pricing Advisor Design

An AI-powered dynamic pricing advisor was designed with the following rule-based engine (foundation for LLM enhancement):

Signal	Rule	Action
Occupancy >70% + below neighbourhood median	Price undervalued	Increase 10-15%
Occupancy <20% + 60+ days available	Price may be too high	Decrease 10%
Jul-Sep approaching	Peak season	Apply 15-25% seasonal premium
Weekend (Fri-Sat)	Weekend premium opportunity	Apply 5-10% surcharge

2. Responsible AI Framework

For deploying ML models in an Airbnb-like rental marketplace:

Principle	Implementation
Transparency	All price recommendations explained in plain language; model confidence scores disclosed; hosts can override
Fairness	Regular bias audits across neighbourhoods and host types; separate models for different market segments; no protected characteristics in feature set
Privacy	Guest data anonymized before model training; no individual guest profiling; PDPA compliant
Accountability	Human review required for >20% price change recommendations; audit trail for all model decisions
Safety	Price floor/ceiling guardrails ($\pm 50\%$ from market median); anomaly detection for unusual pricing patterns; rollback mechanism if model performance degrades

3. MLOps Workflow

Continuous model retraining pipeline design:



9.5 Stream Processing Simulation

A Kafka-style real-time price monitoring simulation was implemented (pipeline/stream_processing_sim.py) demonstrating:

Alert Rules:

1. Price spike >50%: HIGH severity alert
2. Price drop >40%: MEDIUM severity alert
3. Price >3x neighbourhood median: LOW severity alert
4. Neighbourhood trend (sustained >20% avg change): INFO alert

Simulation results (50 events): Total alerts fired: 7

Alert types: ABOVE_MARKET (7) - listings priced 3-5.6x above neighbourhood median detected

Production architecture:

Airbnb scraper → Kafka topic → Flink consumer → Alert rules engine → SNS/PagerDuty → TimescaleDB → Grafana dashboards

Scalability: Kafka partitioned by neighbourhood (50 partitions); handles 10,000+ events/second at 50-city scale.

10 Visualizations

All visualizations are available in two formats:

- Static figures: notebooks/figures/ (30 figures, PNG format)
- Interactive: dashboard/app.py (Streamlit dashboard)
- Deployed: airbnb-market-intelligence-primeshmarasinghe.streamlit.app

Figure Index:

Figure	Filename	Description
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11 Business Recommendations

11.1 For Hosts

Recommendation 1: Pursue Superhost Status

Evidence: Superhosts earn 340% more annual revenue (฿114,431 vs ฿25,968) without charging meaningfully higher prices. Cohen's d=0.54 rating advantage

Action: Maintain >90% response rate, achieve 4.8+ rating, minimize cancellations, and ensure consistent availability. Focus on guest communication and cleanliness as highest-impact improvements.

Recommendation 2: Implement Weekend Pricing Premium

Evidence: H5 confirmed zero weekend premium in Bangkok (p=0.526, d=0.012). No competitive norm exists.

Action: Implement 10-15% Friday-Saturday surcharge. Test incrementally - monitor occupancy impact over 4-week periods before increasing further. Early adopters have first-mover advantage before market normalizes.

Recommendation 3: Optimize Listing for BTS Proximity

Evidence: BTS mentioned in 2,513 reviews - far exceeding any other location signal. NER confirms it is the #1 booking decision factor.

Action: Include specific BTS station name and walking time in listing title. "3 min walk to Ekkamai BTS" consistently outperforms generic "great location" descriptions in Bangkok market.

Recommendation 4: Strategic Amenity Investment

Evidence: TV (+68.6%), Hot Tub (+49.8%), Kitchen (+36.9%) command highest price premiums. WiFi (95.3% prevalence, +14.1%) is now table stakes.

Action: Prioritize kitchen installation for entire-home listings -highest ROI amenity for properties lacking one. Hot tub for luxury segment. Avoid competing on WiFi speed alone.

Recommendation 5: Dynamic Pricing for Jul-Sep Peak

Evidence: Occupancy jumps from 24% (Feb trough) to 43% (Sep peak) - a 79% demand swing within the same year.

Action: Implement 15-25% price increase for July-September bookings. Use early-bird discounts (10%) for October-March trough period to stimulate advance bookings.

11.2 For Investors**Recommendation 6: Target Bang Rak and Vadhana**

Evidence: Bang Rak commands highest median price (฿1,875) with strong demand.

Vadhana offers deepest market (4,305 listings) with high liquidity for entry and exit.

Action: Bang Rak for premium positioning and pricing power. Vadhana for portfolio scale and market liquidity. Avoid outer suburbs despite higher ratings - insufficient demand volume.

Recommendation 7: Focus on Mid-Market Entire Homes

Evidence: ฿1,000 - 2,500 price range captures highest demand concentration. High Demand segment achieves 61% occupancy at ฿1,411 median.

Action: Target entire-home apartments in Sukhumvit corridor (Vadhana, Khlong Toei, Ratchathewi) priced ฿1,200-2,000. This profile optimizes the occupancy-revenue trade-off.

Recommendation 8: Bangkok Market Fundamentals Are Strong

Evidence: 2024 reviews at all-time high (+152% vs 2019). Market has not merely recovered from COVID - it has structurally grown.

Action: Continue investment in Bangkok short-term rental assets. Thailand's ongoing tourism promotion and infrastructure investment support continued demand growth through 2026-2028.

11.3 For Platform Operators

Recommendation 9: Monitor Commercial Host Concentration

Evidence: 10.6% of hosts control 57% of listings. Top host manages 243 listings. Bangkok structure mirrors pre-regulation Amsterdam and Barcelona.

Action: Establish listing count thresholds for commercial operators. Proactively engage Thai regulatory authorities before legislation is imposed reactively. Distinguish casual hosts (policy allies) from commercial operators (policy targets).

Recommendation 10: Address Rating Inflation

Evidence: 81.4% of listings rated above 4.5. Zero listings with >50 reviews and <4.0 rating. Sub-dimension scores compressed in 4.65-4.78 range.

Action: Promote sub-dimension scores (cleanliness, location, value, communication) as primary quality indicators. Implement "verified quality" badges based on sub-dimension performance rather than overall rating. Consider resetting overall ratings periodically to prevent compression.

Recommendation 11: Multilingual Review Analysis

Evidence: 40% of Bangkok reviews are non-English (German, Spanish, French confirmed by LDA topic modeling). VADER sentiment underestimates negative signals from non-English reviews.

Action: Deploy XLM-RoBERTa for multilingual sentiment analysis. Implement language detection pipeline. Non-English speaking guests represent high-value segments whose feedback is currently invisible to the platform's NLP systems.

12 Cross-City Comparisons

Single-city analysis was conducted for this assessment. The following outlines how cross-city comparison would be approached with additional time.

Planned Comparison Markets:

City	Rationale
Singapore	Regional premium market benchmark
Kuala Lumpur	Similar SEA tourist profile
Ho Chi Minh City	Emerging competitor market
Bali	Resort-focused alternative

Harmonization Strategy:

1. Currency normalization: All prices to USD at scrape-date exchange rate
2. Neighbourhood taxonomy: Admin level 2 for all cities
3. Scrape date alignment: Within 90-day window for comparability
4. Population normalization: Listing density per 100K residents

Expected findings:

- Bangkok expected to show lower price points than Singapore but higher than Ho Chi Minh City
- Commercial host concentration likely highest in Bangkok vs regional peers
- Seasonal patterns expected to diverge significantly - Bali peaks in Australian school holidays vs Bangkok's European summer

13 Limitations & Caveats

13.1 Data Limitations

Limitation	Impact	Severity
Calendar prices null	Revenue estimates are proxies only	High — primary revenue metric is approximate
Occupancy proxy (availability_365)	Conflates host blocking with bookings	Medium — may overestimate occupancy by 10-20%
Single scrape date	No longitudinal price tracking	Medium — cannot track price changes over time
Review scores 35% missing	Ratings analysis on subset only	Medium — subset only; may bias toward active

		listings
No transaction data	Cannot verify actual bookings	High — listed price $\neq$ transaction price
Multilingual reviews	NLP misses 40% of content	Medium — English-only VADER underestimates non-English sentiment

13.2 Methodological Limitations

Occupancy Proxy: The availability_365 field records days marked unavailable by hosts, which may reflect bookings OR deliberate blocking. Inside Airbnb acknowledges this limitation in their documentation. True occupancy is likely 10-20% lower than estimated. All occupancy figures should be interpreted as upper bounds.

Revenue Estimation: Using listed price $\times$ estimated booked days overstates revenue for listings offering length-of-stay discounts and understates for listings with dynamic pricing premiums. Median revenue (£61,848) should be treated as an order-of-magnitude estimate, not a precise figure.

Causality: All findings are correlational. Superhost status correlating with higher ratings does not establish that achieving superhost status causes rating improvement - selection effects are present (higher-quality hosts may both achieve superhost status and receive higher ratings independently).

Temporal Validity: The September 2025 scrape reflects a single point in time. Market conditions, pricing strategies, and competitive dynamics evolve continuously. Findings should be reviewed against updated scrapes quarterly.

Statistical Limitations:

- Large sample sizes ($n > 8,000$) make even trivial differences statistically significant effect sizes must be interpreted carefully throughout
- Multiple hypothesis testing without Bonferroni correction - familywise error rate is elevated across 5 tests
- Non-independence of listings from same host violates independence assumptions of standard tests

Model Limitations:

- Price model trained on listed price not transaction price - may not reflect actual market clearing prices
- Geographic bias confirmed - model underperforms in premium areas

- Luxury segment (>¥10,000) excluded from training - model should not be applied to this segment
- Model requires retraining as market conditions evolve

14 Future Improvements

14.1 With More Time

Improvement	Value Added
Multi-city comparative analysis	Regional benchmarking and cross-market insight
Transformer-based NLP (XLM-RoBERTa) SHAP explainability	Multilingual sentiment accuracy; semantic topic modeling Per-prediction feature importance; interaction effect detection
Geospatial clustering (GeoPandas + Folium) Great Expectations	Neighbourhood boundary visualization; distance-to-center analysis Automated data quality gates; HTML data documentation
dbt execution	Full SQL-first transformation layer; auto-generated lineage graphs
Longitudinal analysis (multiple scrapes)	12+ month price trend tracking; true seasonality modeling

14.2 With Better Data

Data Gap	What It Would Enable
Actual transaction prices	True revenue modeling and pricing efficiency
Booking timestamps	Real occupancy calculation (not proxy)
Guest demographics	Demand segmentation by traveler profile
Host cost data	True profitability analysis
View/click data	Recommendation system collaborative filtering
Historical scrapes	Longitudinal price trend analysis
Platform search rankings	Visibility and conversion rate analysis

14.3 With Production Infrastructure

Component	Current State	Production Recommendation
Orchestration	Manual scripts	Apache Airflow DAG (weekly)
Storage	Local CSV files	AWS S3 data lake + Glue catalog
Processing	pandas (local)	Spark for multi-city parallel
Serving	Local Streamlit	QuickSight + API Gateway
Monitoring	File logging	CloudWatch + PagerDuty alerting
CI/CD	Manual push	GitHub Actions automated testing
Data Quality	pytest (21 tests)	Great Expectations + Slack alerts
ML Serving	Offline batch	SageMaker real-time endpoint

15 Reflection

15.1 Prioritization Decisions

The one-week timeline required deliberate prioritization. The core data engineering pipeline was completed first as the foundation for all downstream work. EDA and hypothesis testing followed as the primary analytical deliverables given their high rubric weights (25 and 20 points respectively). The Streamlit dashboard and AI Analyst were built as open innovation components demonstrating product thinking.

Sections completed in priority order:

1. Data pipeline (ingestion → star schema) - core competency signal
2. EDA with business interpretations - storytelling signal
3. Hypothesis testing with effect sizes - statistical rigor signal
4. ML price prediction with cross-validation - data science signal
5. NLP sentiment + topic modeling + NER - AI literacy signal
6. Dashboard + AI Analyst + automated report - creativity signal
7. dbt models + Docker + unit tests - professionalism signal

15.2 Key Trade-offs Accepted

Trade-off	Decision	Rationale
Depth vs breadth	Single city, deep analysis	Assignment explicitly rewards depth
Multi-city vs ML	ML modeling priority	Higher rubric weight; more impactful analysis
Cloud vs local	Local pipeline, cloud design	Reproducibility without cloud costs or complexity
Speed vs quality	Slower, documented	Report quality is primary evaluation artifact
SHAP vs feature imp.	Built-in importance	NumPy version conflict; equivalent directional insight

15.3 Key Lessons Learned

1. Data quality issues surface early or not at all: The null calendar prices were discovered during profiling, allowing time to design a principled workaround. Running `profile.py` before any analysis is non-negotiable in production data engineering.
2. Effect sizes matter as much as p-values: H3 demonstrated that statistical significance without practical significance is misleading. At $n=23,039$, even a \$41 difference achieves $p<0.00001$. Always report Cohen's d or eta-squared alongside p-values.
3. Commercial concentration is a Bangkok-specific story: The 57% commercial operator finding was not anticipated and emerged from the data, representing genuine discovery rather than hypothesis confirmation. This is the most actionable platform-level insight in the report.
4. Documentation is part of the engineering: Maintaining decision logs, assumption documentation, and metadata throughout the pipeline made the final report significantly easier to write and substantially more credible to reviewers.
5. NLP reveals what quantitative data cannot: The LDA finding that 40% of reviews are non-English was entirely invisible in the numerical dataset. This finding has direct implications for platform investment in multilingual NLP capabilities.

6. The AI Analyst demonstrates end-to-end product thinking: Building a conversational interface over the analysis data showed that the work can be packaged for non-technical stakeholders - a skill valued in data engineering roles that interface with business teams.

Appendix A - AI Usage Disclosure

A.1 AI Tools Used

Tool	Version	Purpose
Claude (Anthropic)	Claude Sonnet 4.6	Primary coding assistant, report structuring, debugging
GitHub Copilot	-	Inline code completion

A.2 AI-Assisted Sections

Section	Assistance Level	Description
Pipeline code (ingest, clean, enrich)	High	Generated with prompting, reviewed and modified
EDA notebook cells	High	Generated with prompting, all outputs verified
Hypothesis testing code	Medium	Generated; statistical choices independently verified
Dashboard CSS/layout	Medium	Generated; significantly modified for aesthetics
ML pipeline code	Medium	Generated; CV and bias analysis independently designed
NLP pipeline code	Medium	Generated; findings independently interpreted
Report structure	Medium	Structure suggested; all content written and verified
Business interpretations	Low	All independently authored
Statistical conclusions	Low	All independently verified against actual outputs

A.3 Key Prompts Used

All interactions conducted with Claude Sonnet 4.6 via claude.ai.

Representative prompts:

"Build a repeatable ingestion pipeline with retry logic and metadata tracking for Inside Airbnb Bangkok data"

"Clean the listings CSV - standardize price, parse dates, flag invalid records, derive host tenure"

"Run Mann-Whitney U test for H1 with effect size and business interpretation"

"Build a dark-themed Streamlit dashboard with plotly charts for Bangkok Airbnb data"

"Train and compare Ridge, Random Forest, and Gradient Boosting for price prediction with 5-fold cross-validation"

"Apply VADER sentiment analysis and correlate with numerical review scores"

"Build a TF-IDF RAG system over 10,000 Bangkok Airbnb reviews"

A.4 Output Validation

All AI-generated code was:

1. Executed and outputs verified against expected ranges
2. Reviewed for logical correctness before proceeding
3. Modified where outputs were incorrect (calendar price null issue required independent diagnosis and resolution)
4. Statistical results cross-checked against manual calculations
5. Business interpretations independently authored - AI was not asked to interpret findings, only to generate code

A.5 Critical Assessment

Cases where AI suggestions were rejected or substantially modified:

1. Calendar price handling: AI-generated code produced NaN revenue outputs. Independent diagnosis identified the root cause (Bangkok calendar prices entirely null in source data). Workaround designed independently using `availability_365` proxy.
2. SHAP implementation: NumPy version conflict prevented SHAP installation. Independently diagnosed and documented; used built-in `feature_importances_` as equivalent solution.
3. Cross-validation implementation: Initial CV code used incorrect metric conversion (`expm1` applied to fold scores not predictions). Bug identified and fixed independently.
4. Dashboard CSS: Initial generated CSS had specificity conflicts causing styling to not apply. Independently restructured selector hierarchy.
5. LDA topic labels: AI suggested generic English topic labels. Independent inspection of actual top words revealed 3 of 6 topics were non-English - a finding the AI did not identify. Labels corrected to reflect actual content.

A.6 Responsible AI Practice

AI tools were used as productivity multipliers, not replacements for analytical judgment. All findings, interpretations, and recommendations represent the author's independent analysis validated against actual data outputs. The calendar price discovery, geographic bias finding, commercial concentration insight, and multilingual review finding were all identified through independent data exploration rather than AI-generated insights.

Treating AI disclosure as an afterthought is itself a red flag. This appendix documents AI usage with intentionality, critical thinking, and professional accountability - consistent with Expernetic's assessment criteria.